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College of Engineering & Computer Science

Department of Mechanical Engineering

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Abstract

In this report, we are dealing with a 3D bracket with a 3D load acting on it with a welded surface. Moreover, in depth detail of Ansys Apdl is done for the 3D bracket and the verification of the result is compared it together at the end of this report.

Introduction

A bracket is a component in one of the aspects of mechanical engineering that which has a rigid structure that is fixed to a vertical surface and has a horizontal surface that needs to support weight. It uses are usually window box or beams.

Welding a process used in joining of two materials together mostly with a filler metal with a high melting point.

Now a bracket can be considered into a 2D or 3D problem. Furthermore, in this report, the bracket we are dealing with is a 3D problem. If it was only 2D, the bracket could have an element type of a plane stress or plane strain depending on the thickness of the bracket. However, a 3D problem of a bracket in this report is considered as a shell problem.

A shell problem is categorized by the following ways:

- It has a constant thickness
- It is a 3D problem
- It has a 3D load

Problem Statement

- A 3D bracket with dimensions (in inches) is subjected to loading in X-Y-Z plane and we must calculate Von Mises.

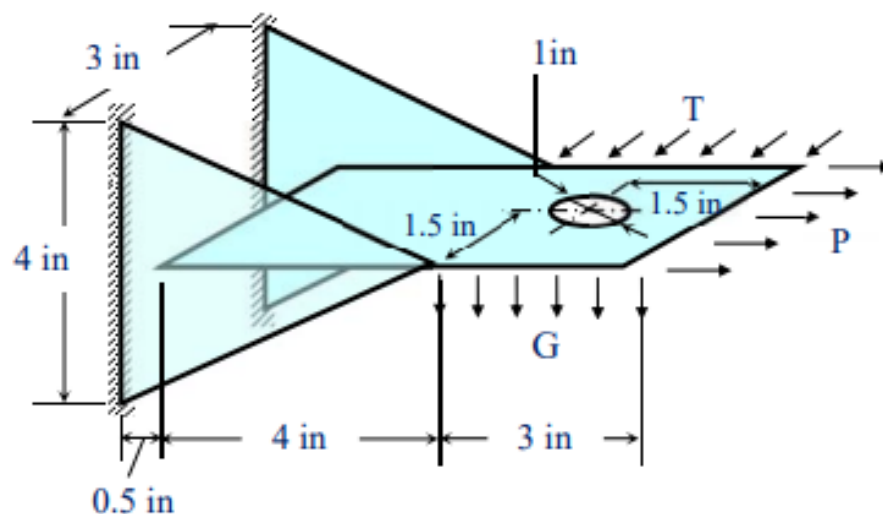


Figure P12

Modeling Plan

- Analysis Type: Static Analysis
- Problem type: Three-Dimensional Shell problem
- Element shape: Quad literal
- Element type:
 - o 6 DOF
 - o 2 DOF/node

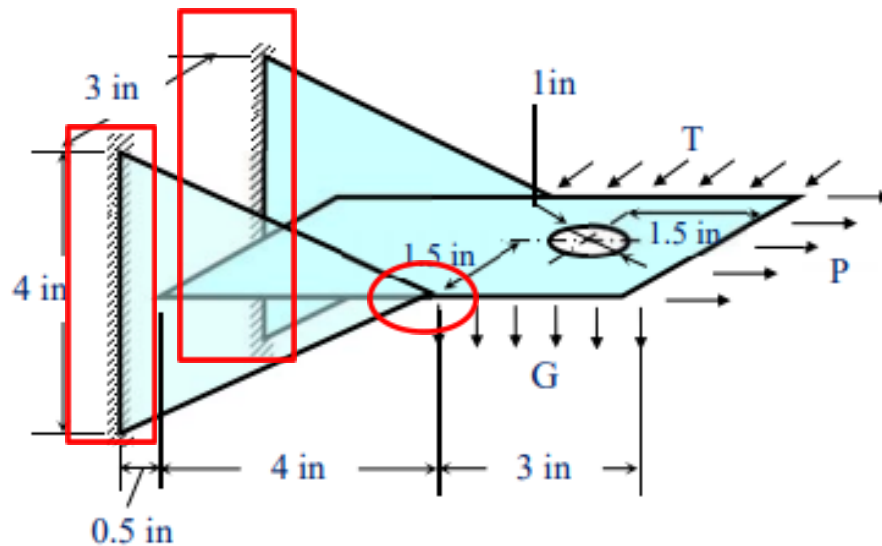


Figure P12

- Red Triangle show that All DOF are equal to zero
- Circle shows where mostly like critical stress is bound to occur due to bending force

Key Points

Key Points	X- Plane	Y-Plane	Z- Plane
1	0	0	0
2	0	4	0
3	0	2	0
4	0.5	2	0
5	0.5	0.2222222	0
6	0.5	3.7777778	0
7	4.5	2	0
8	7.5	2	0
9	7.5	2	3
10	4.5	2	3
11	0	4	3
12	0	0	3

13	0.5	2	3
14	0.5	0.2222222	3
15	0.5	3.7777778	3
16	6	1.5	Radius of 0.5
17	10	0	0
18	0	2	3

Mesh Intensity Plan

- Setting the Global factor to 0.2
- Refining the meshing at the points where there is a circle
- Dividing the lines into more key points for better meshing where the load is being applied
- Line is Divided by N terms for better meshing

Boundary Conditions

- The fixed points at which the bracket is being fixed is equal to zero
- Forces that are being applied towards the lines

Material Properties

The material that which the bracket is made from is Steel with the following properties:

- Modulus of Elasticity = $30 \times 10^9 \text{ psi}$
- Poisson Ratio = 0.3
- Yield Strength = $40 \times 10^3 \text{ psi}$
- Constant Thickness = 0.25
- Pressure = 50 *psi*

Ansys Steps:

Generation of Key Points

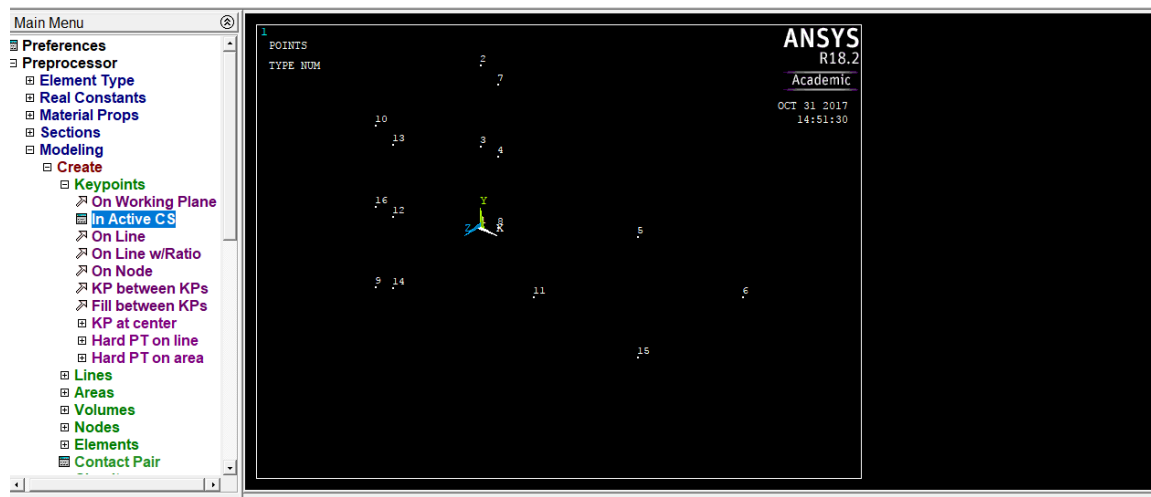


Figure 1 Generation of Key points

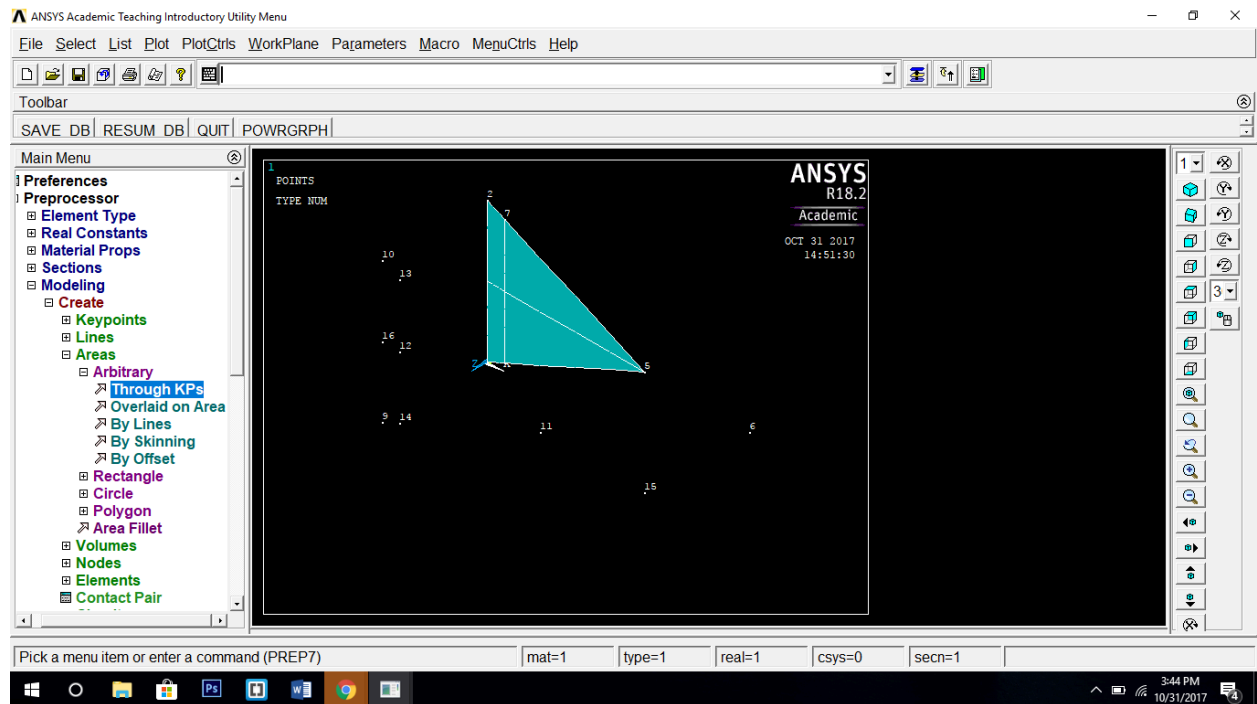


Figure 2 Making of Area 1

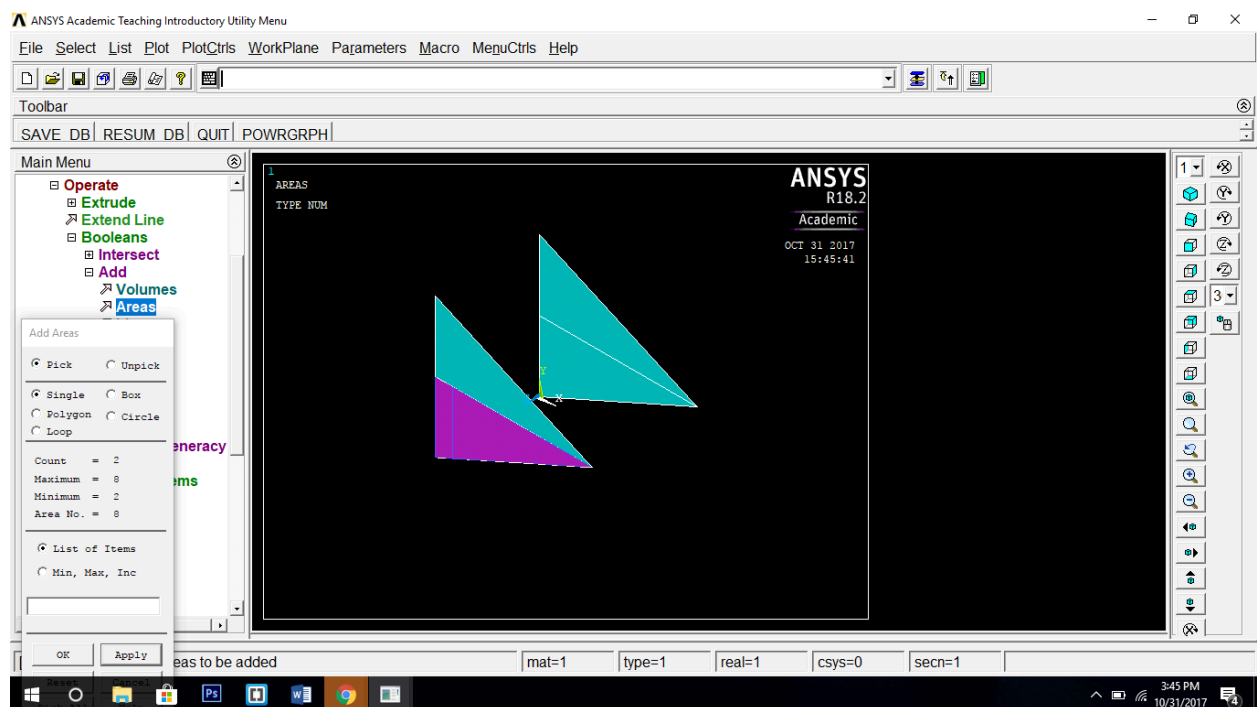


Figure 3 Making of Area 2

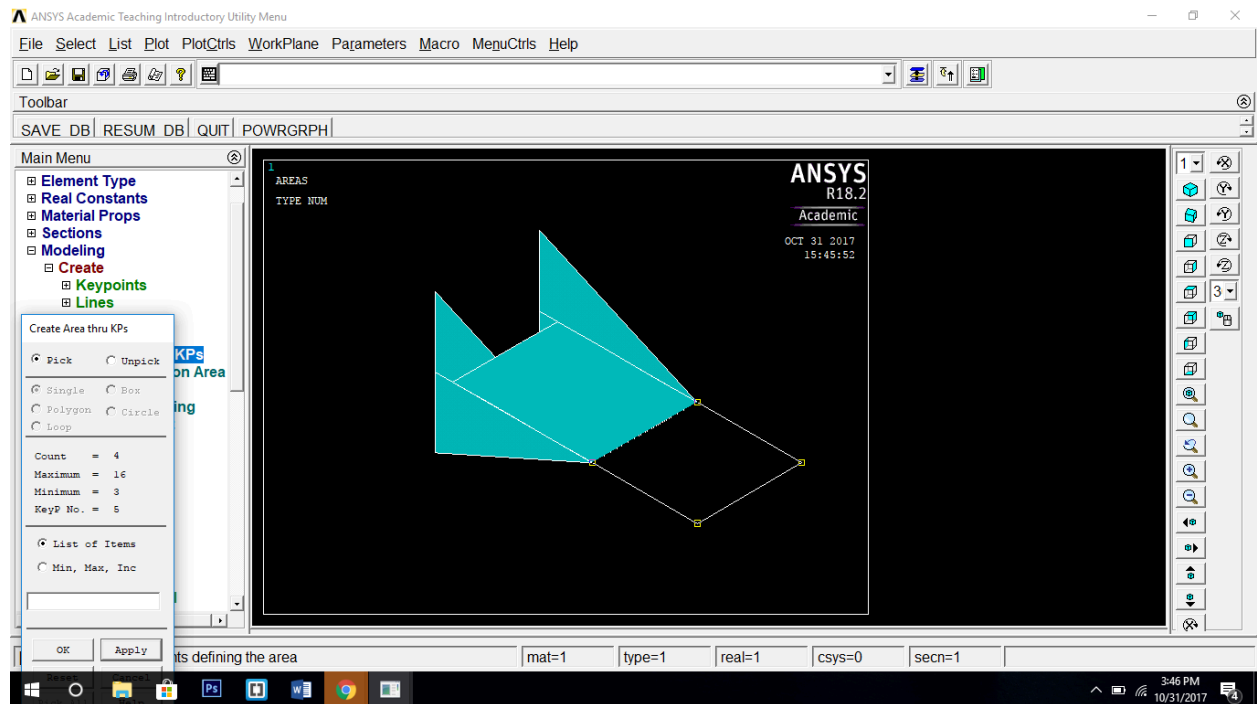


Figure 4 Making of Area of Rectangle

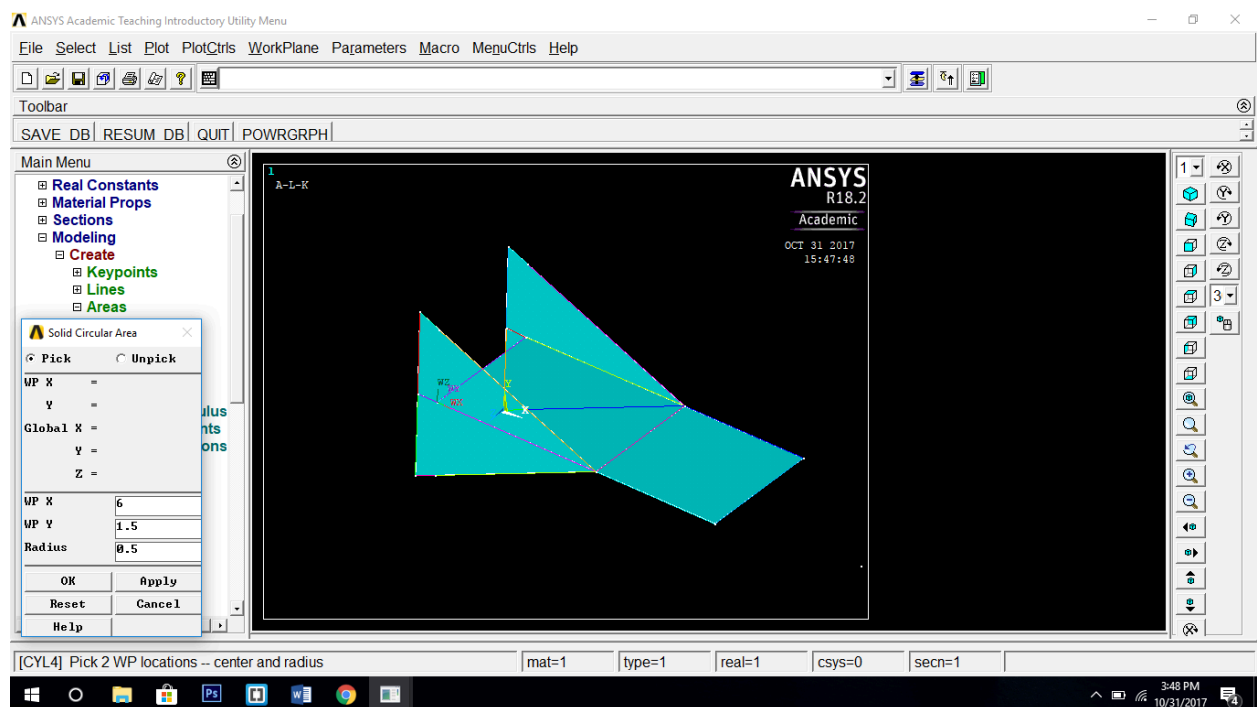


Figure 5 Full Area Completion with welded Areas

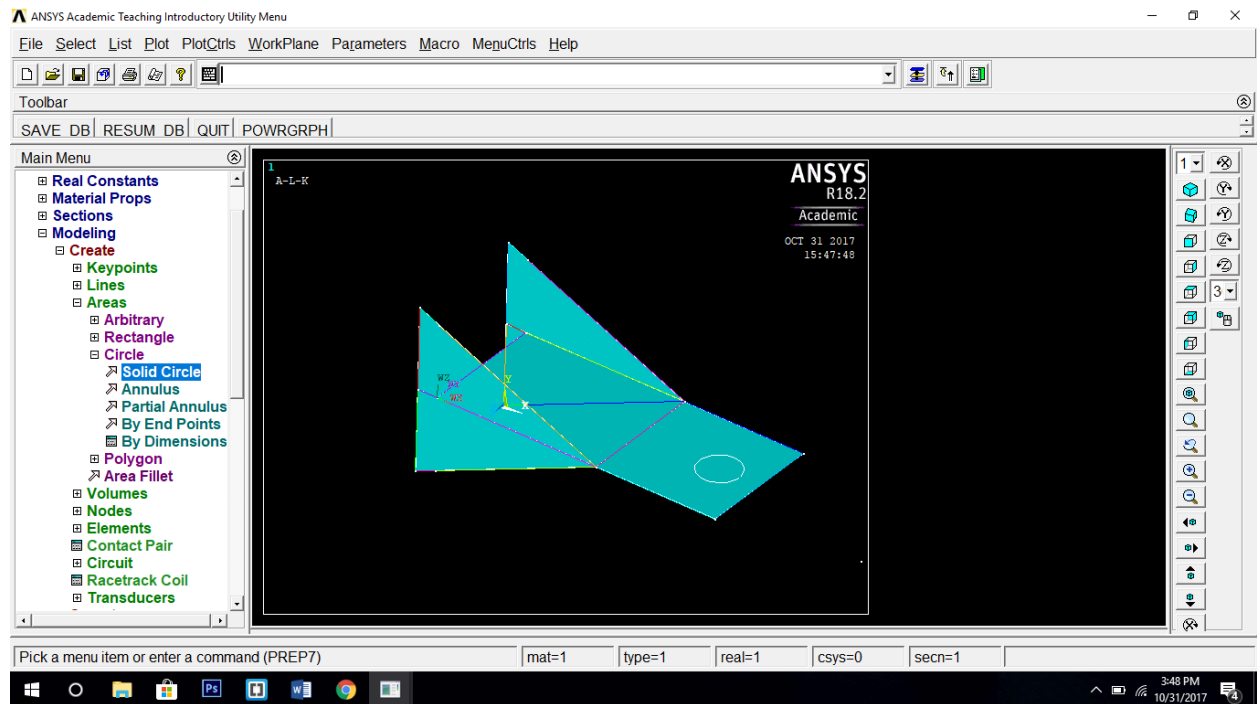


Figure 6 Making a Solid Circle

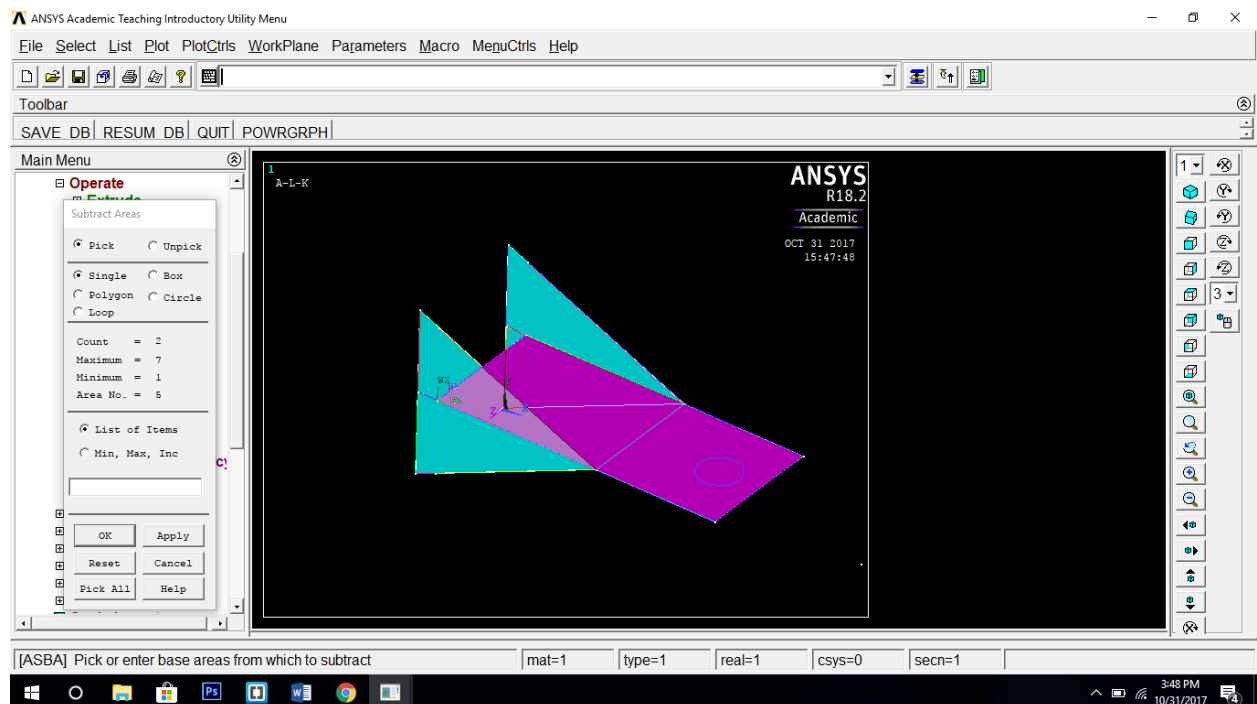


Figure 7 Subtraction 1

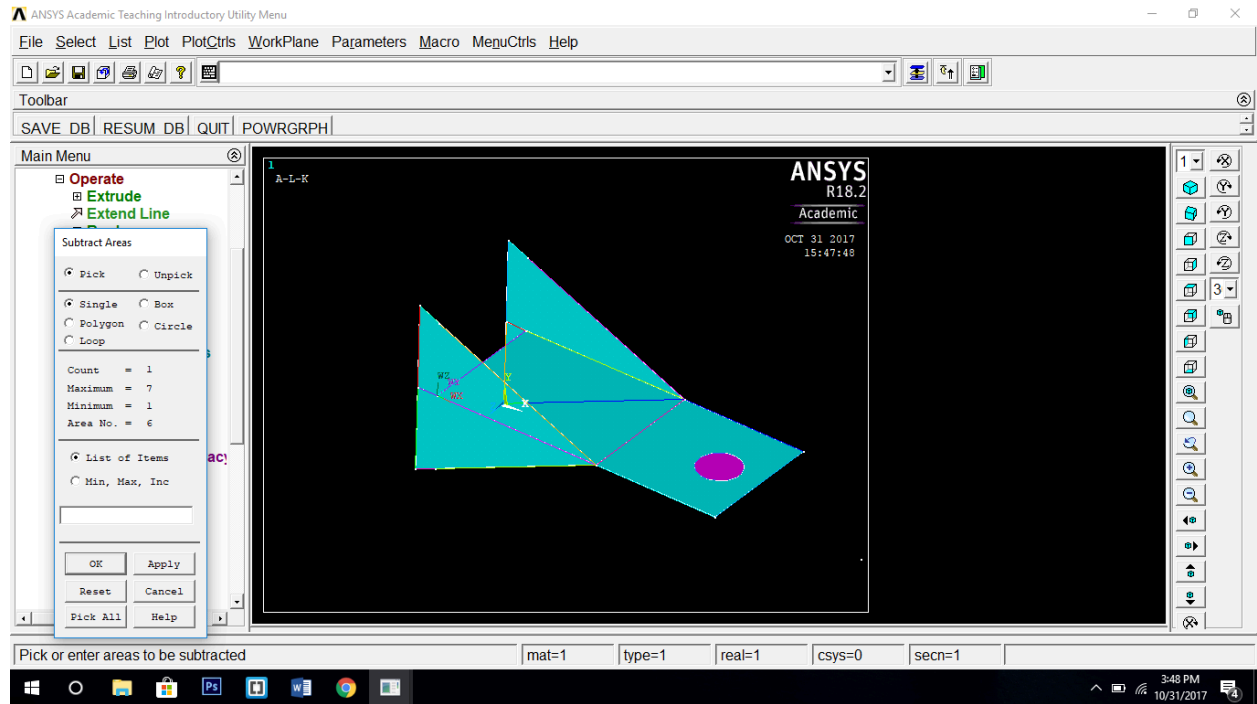


Figure 8 Subtraction (2) Boolean Operation

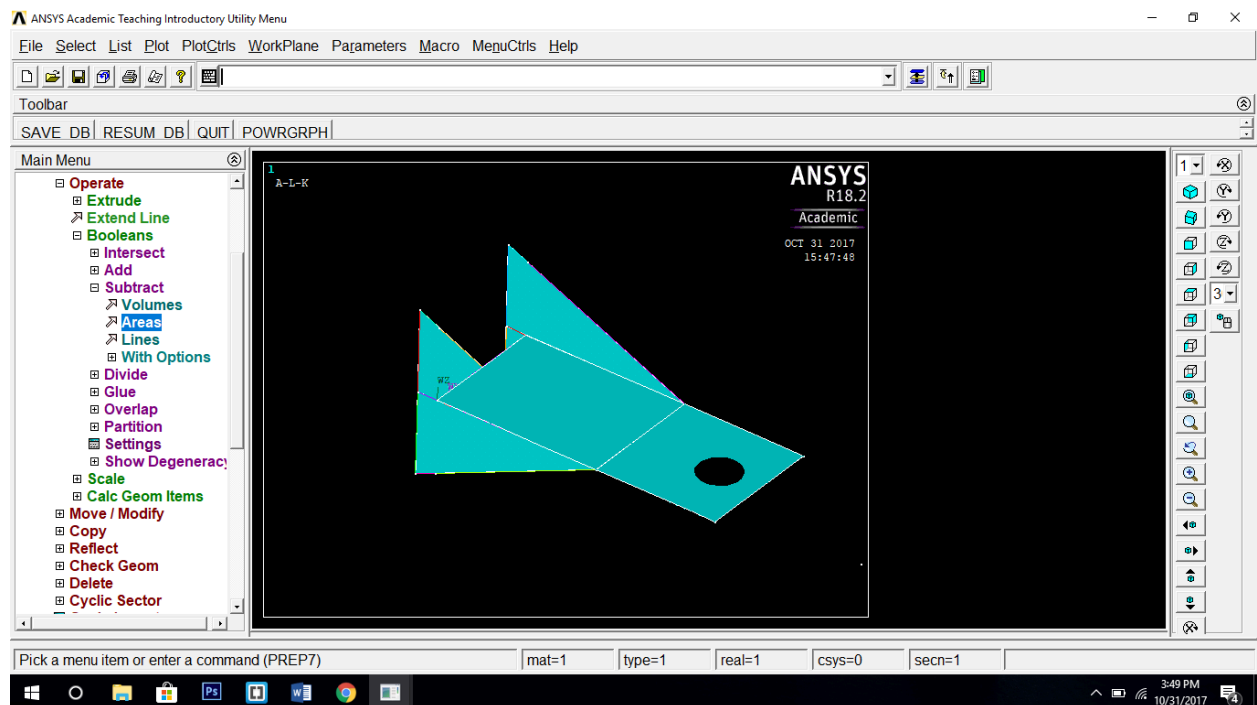


Figure 9 Full Area Completed

Defining the Element type and DOF

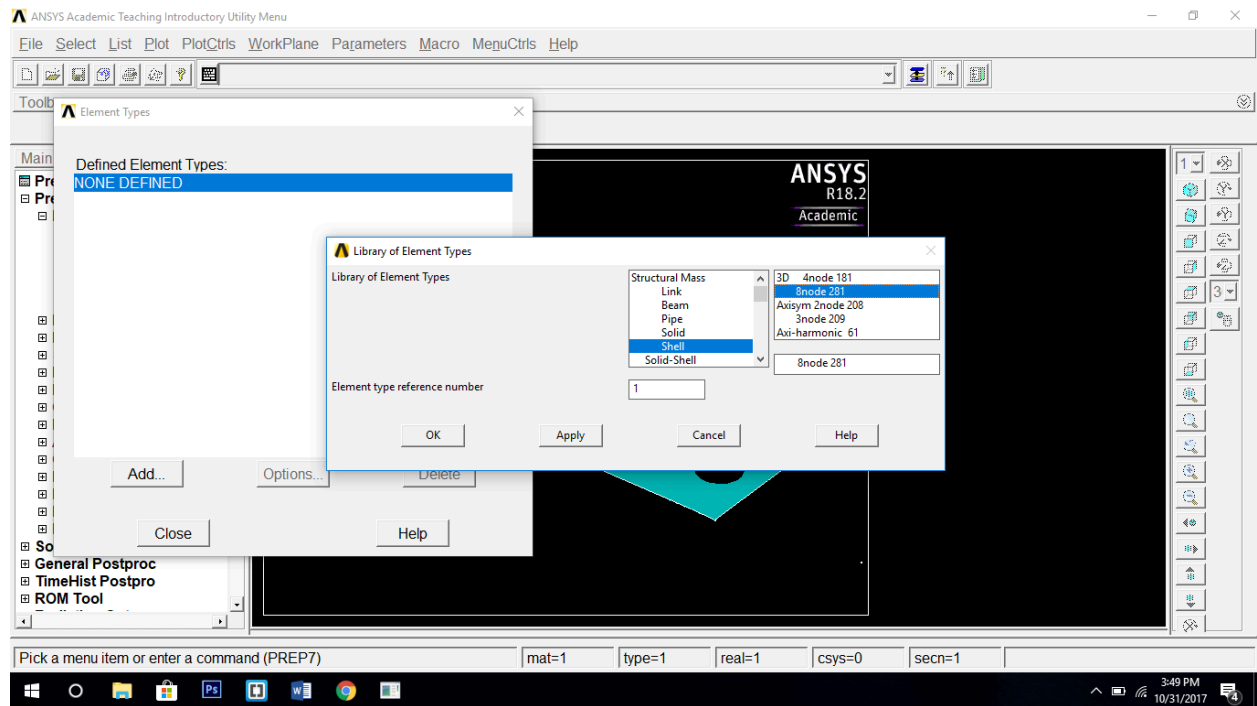


Figure 10 Defining Element Type

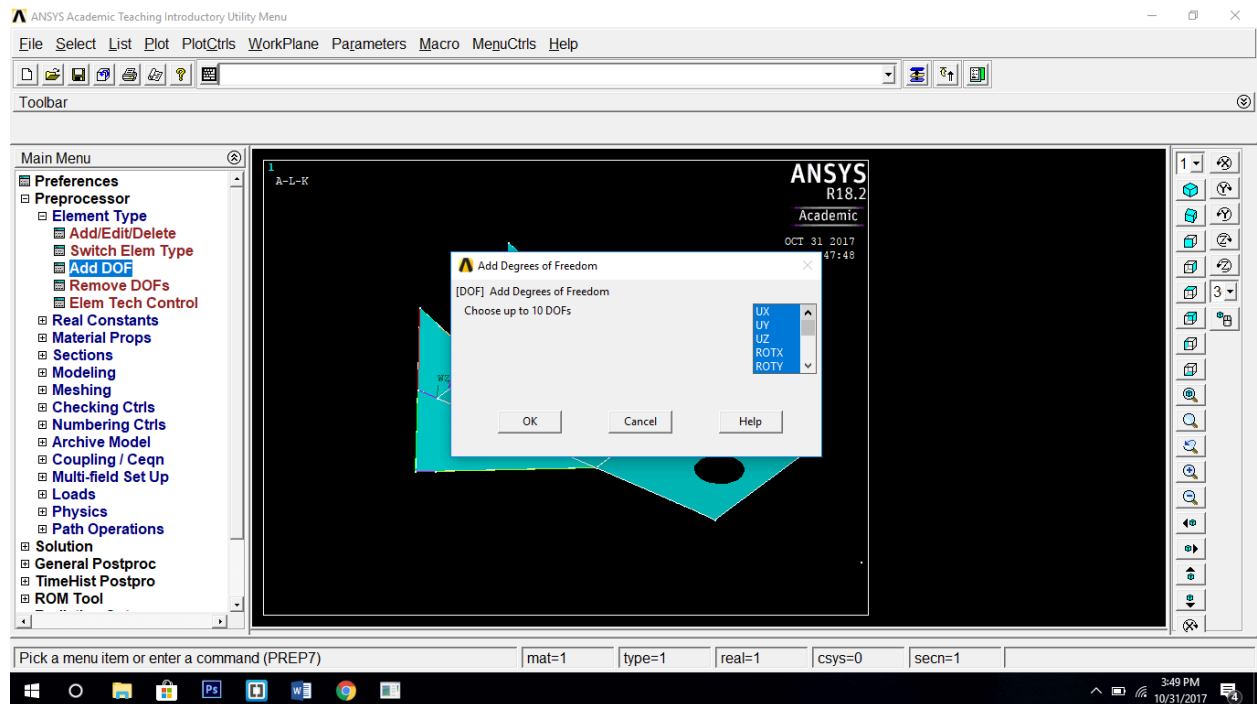


Figure 11 Adding Degree of Freedom

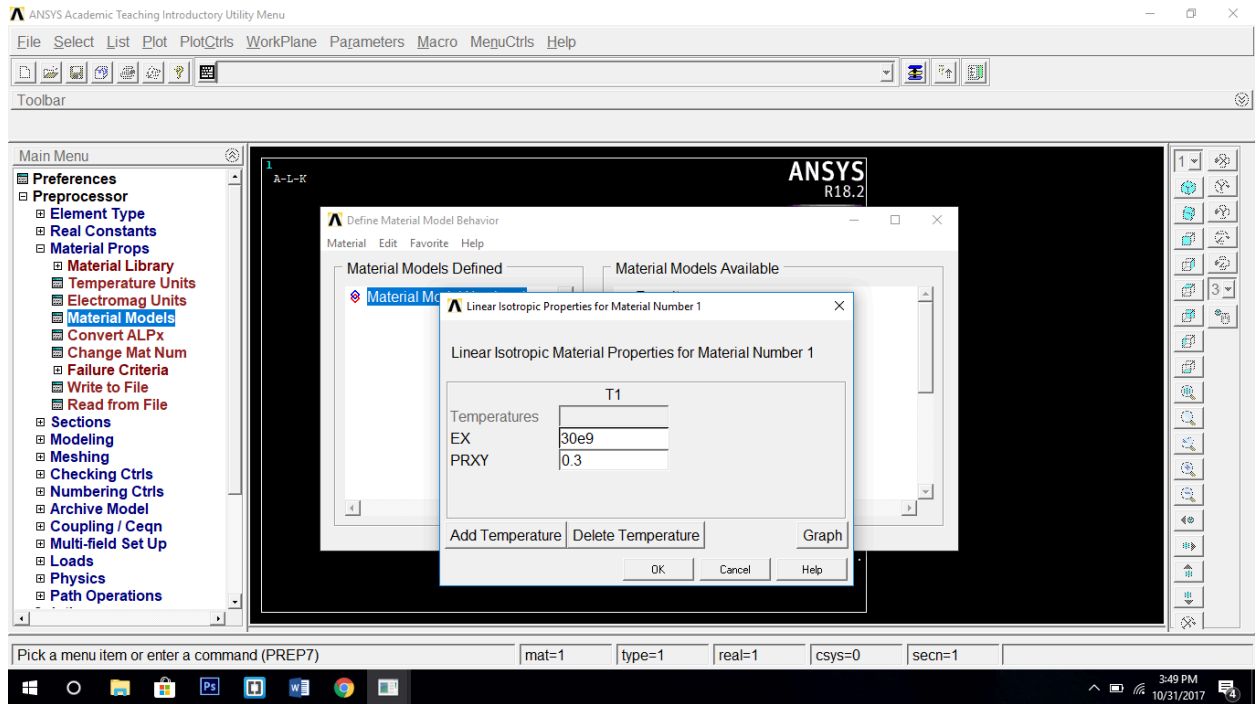


Figure 12 Adding Material Properties

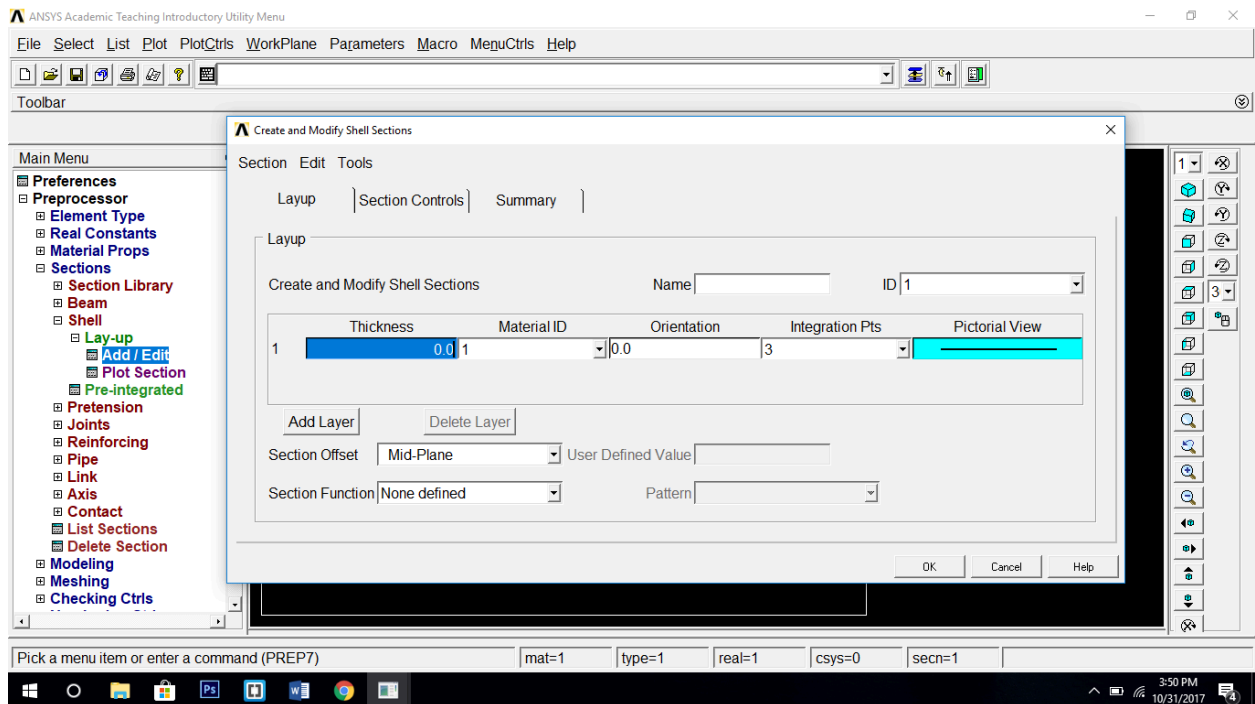


Figure 13 Adding Thickness to the element

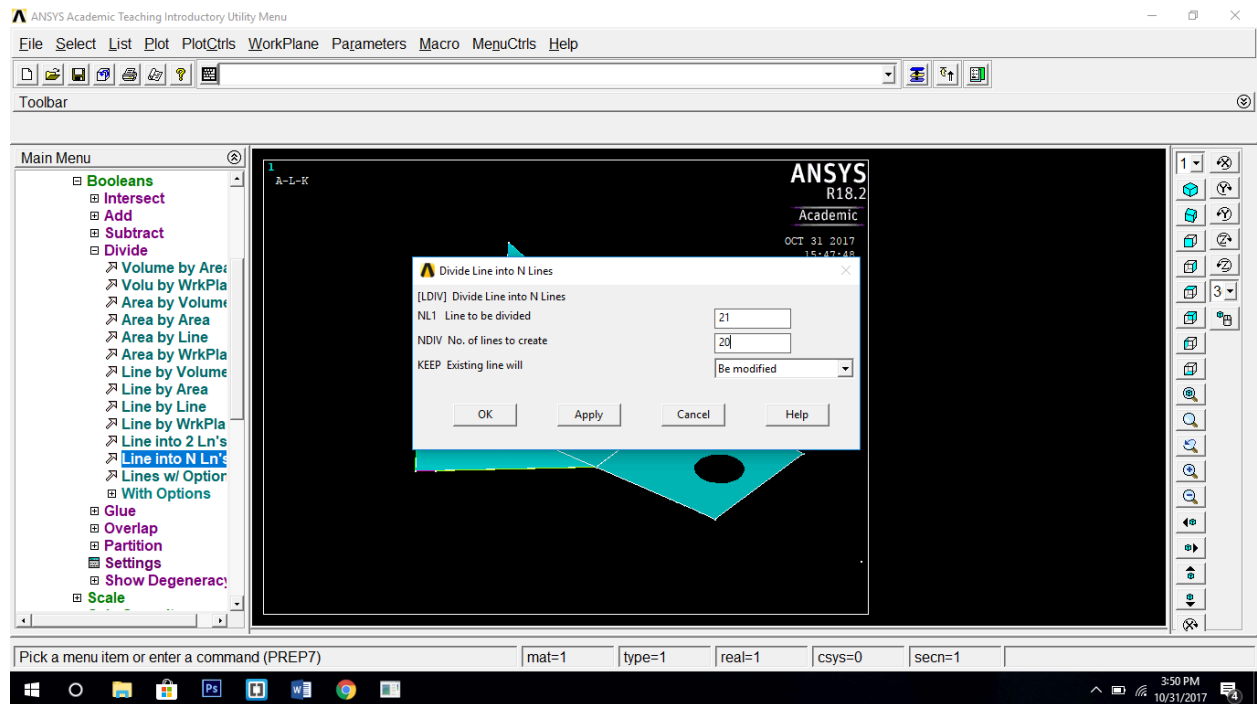


Figure 14 Generating Key points by Divide Operation

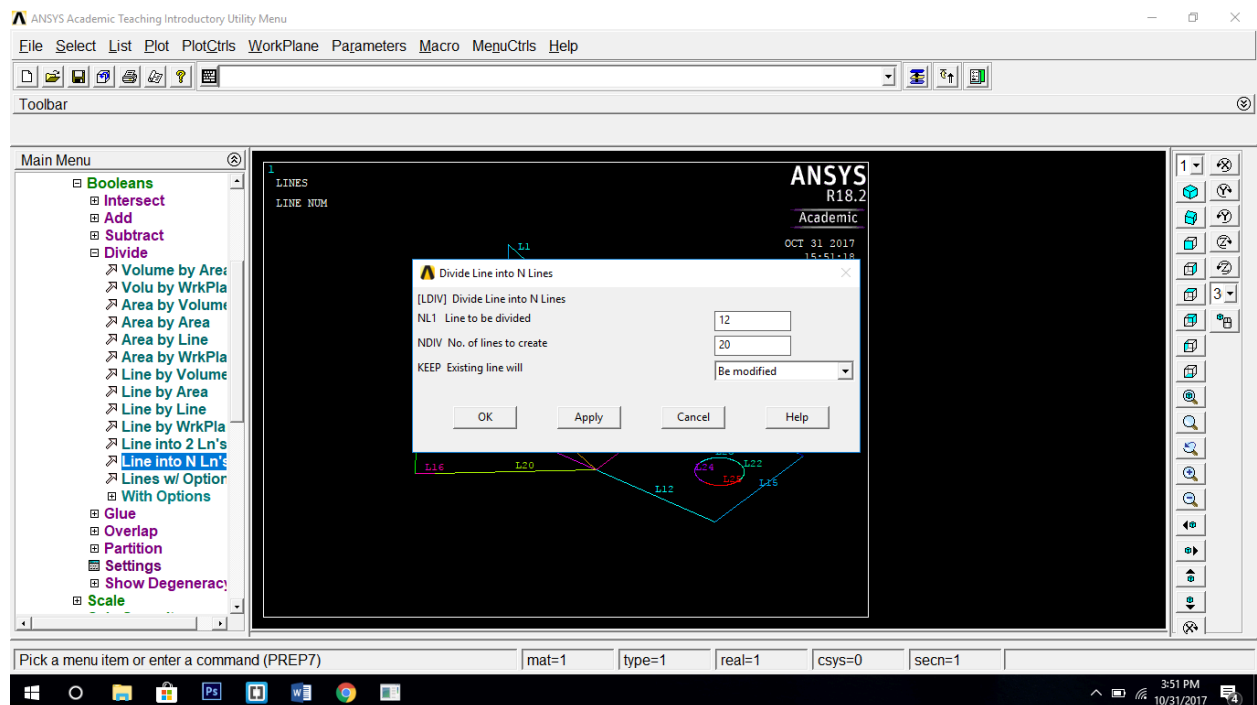


Figure 15 Division of Lines into more Key points

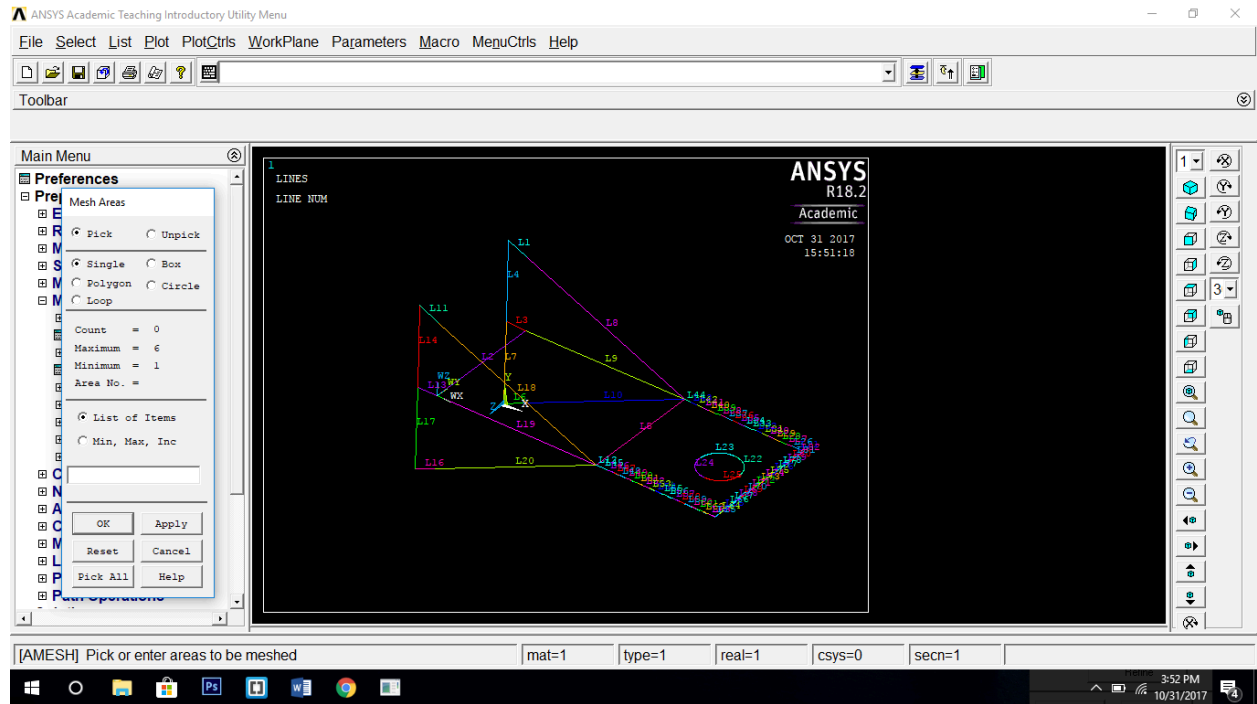


Figure 16 The total view of Key points Division

Meshing of the Element

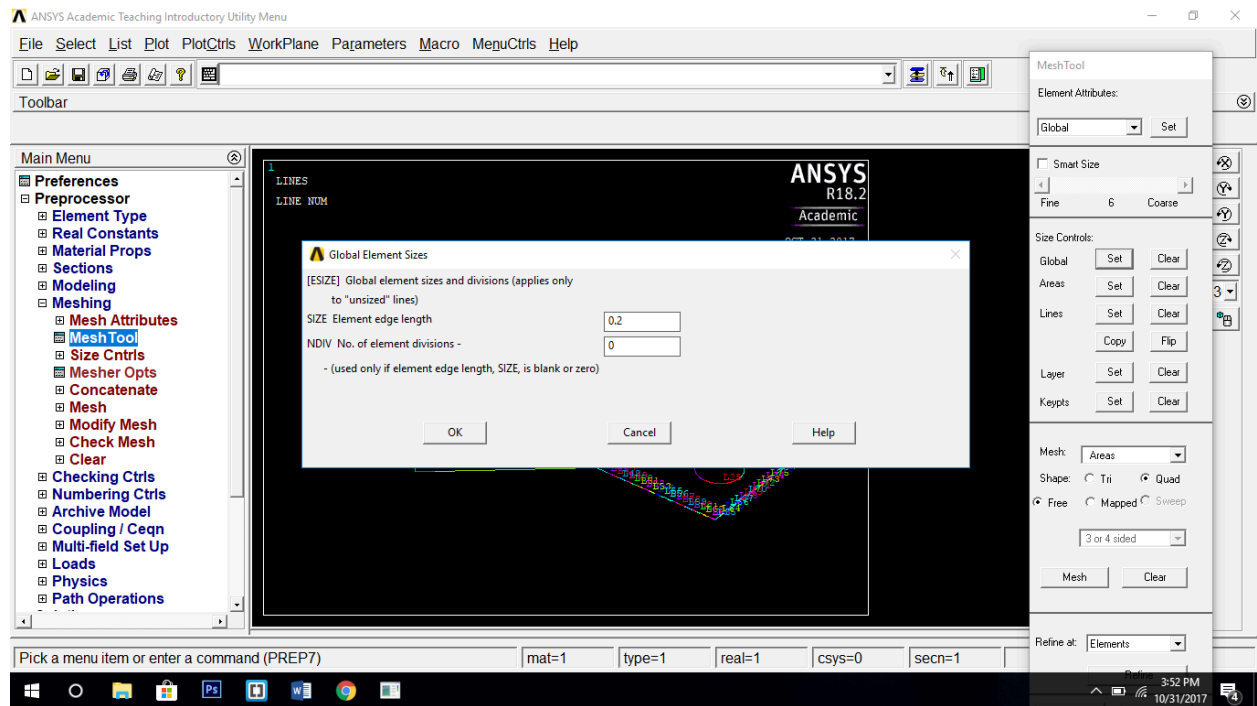


Figure 17 Meshing the Element by a Global factor of edge length

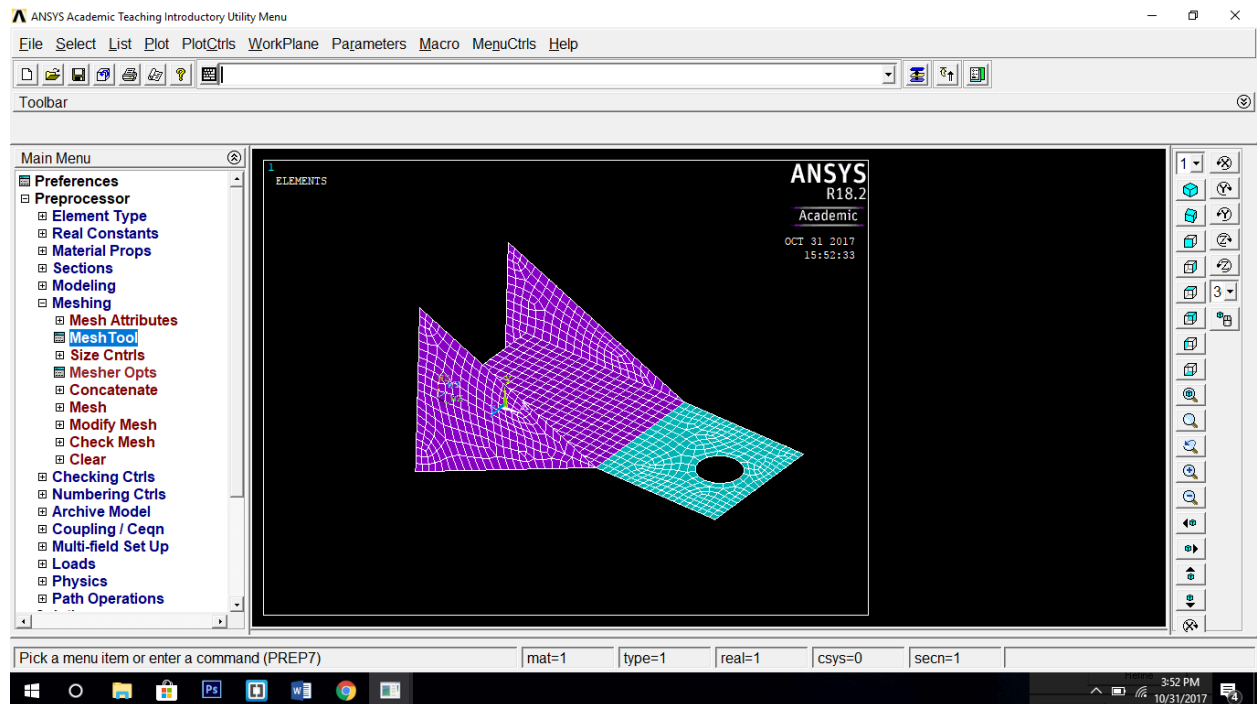


Figure 18 Initial Mesh of the Element

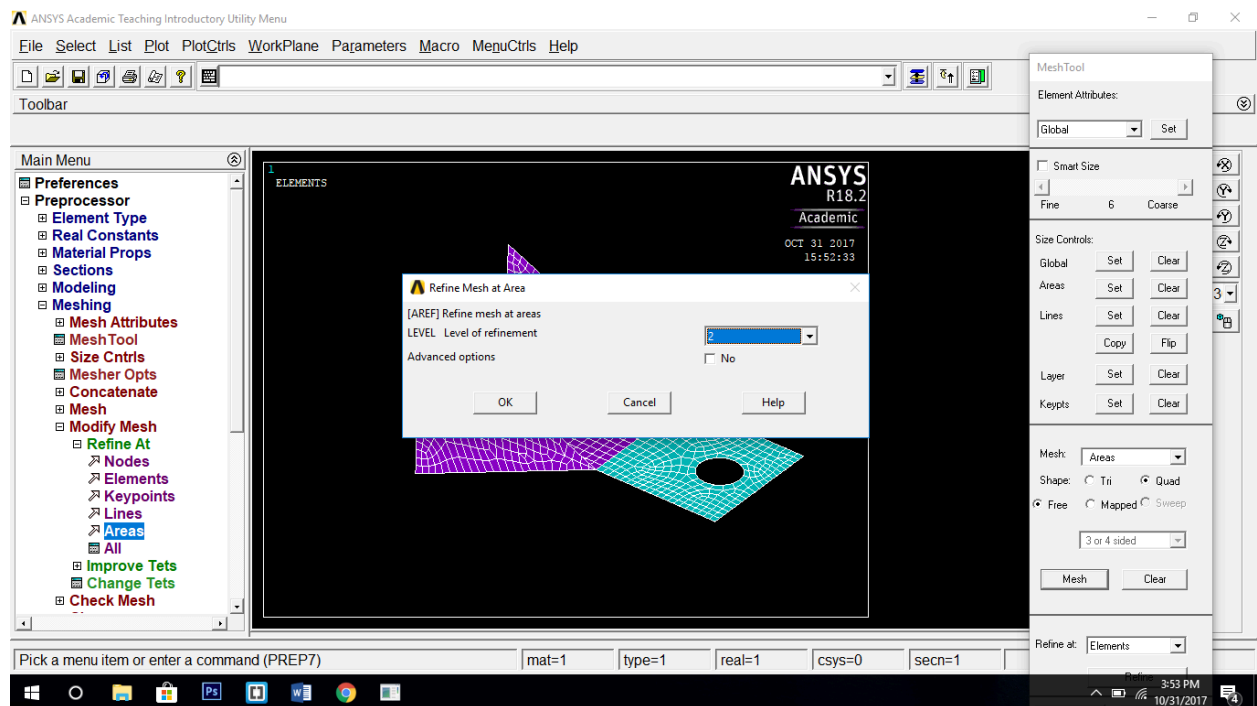


Figure 19 Refining of Mesh/ Modification

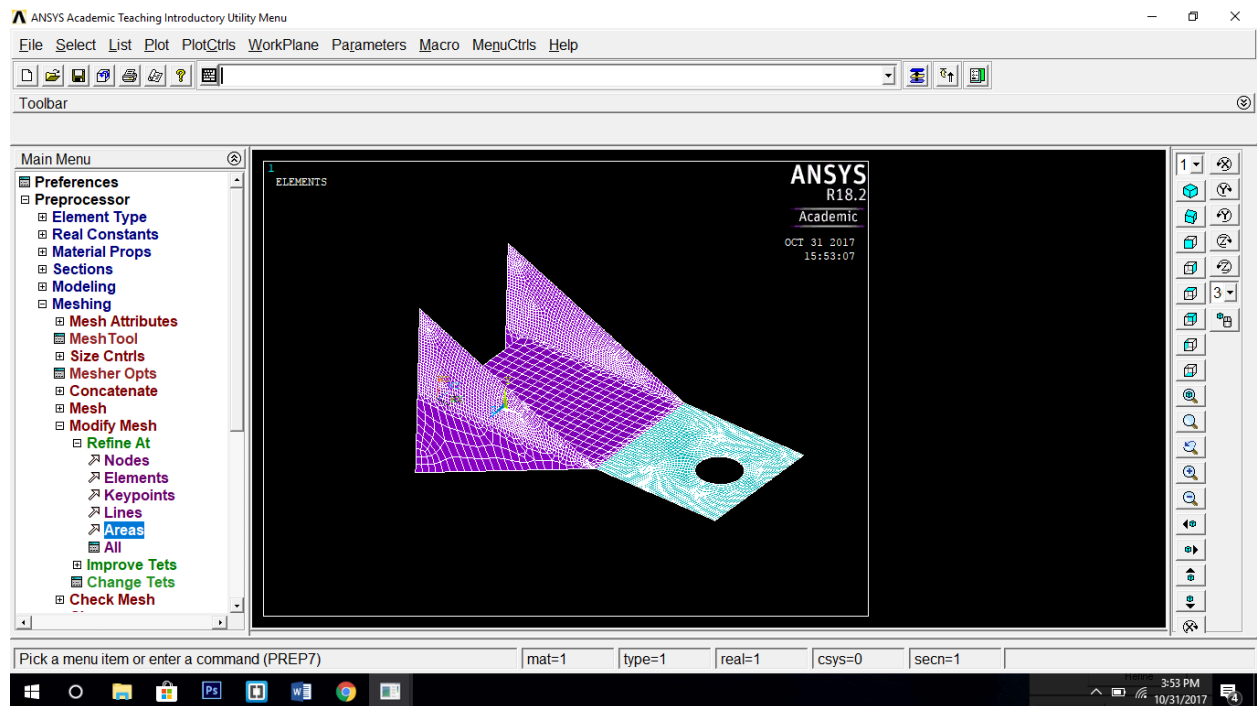


Figure 20 Refining at those area with high stress

Solving

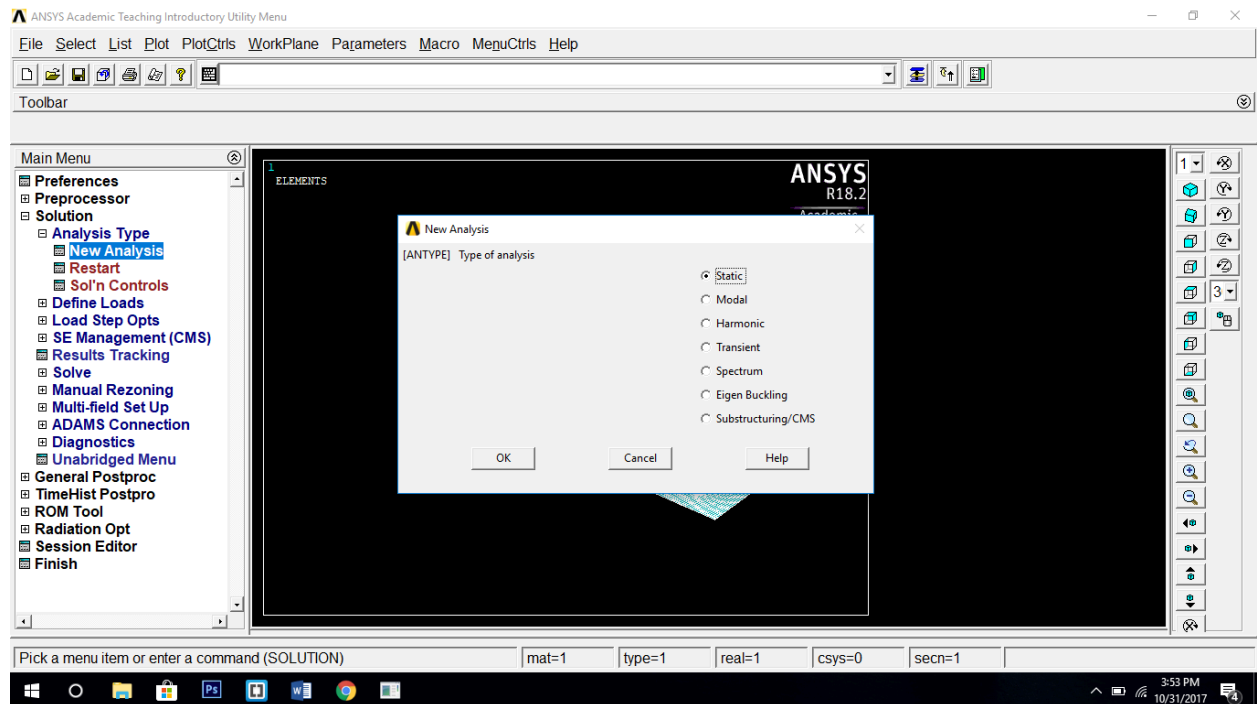


Figure 21 Analysis of Element

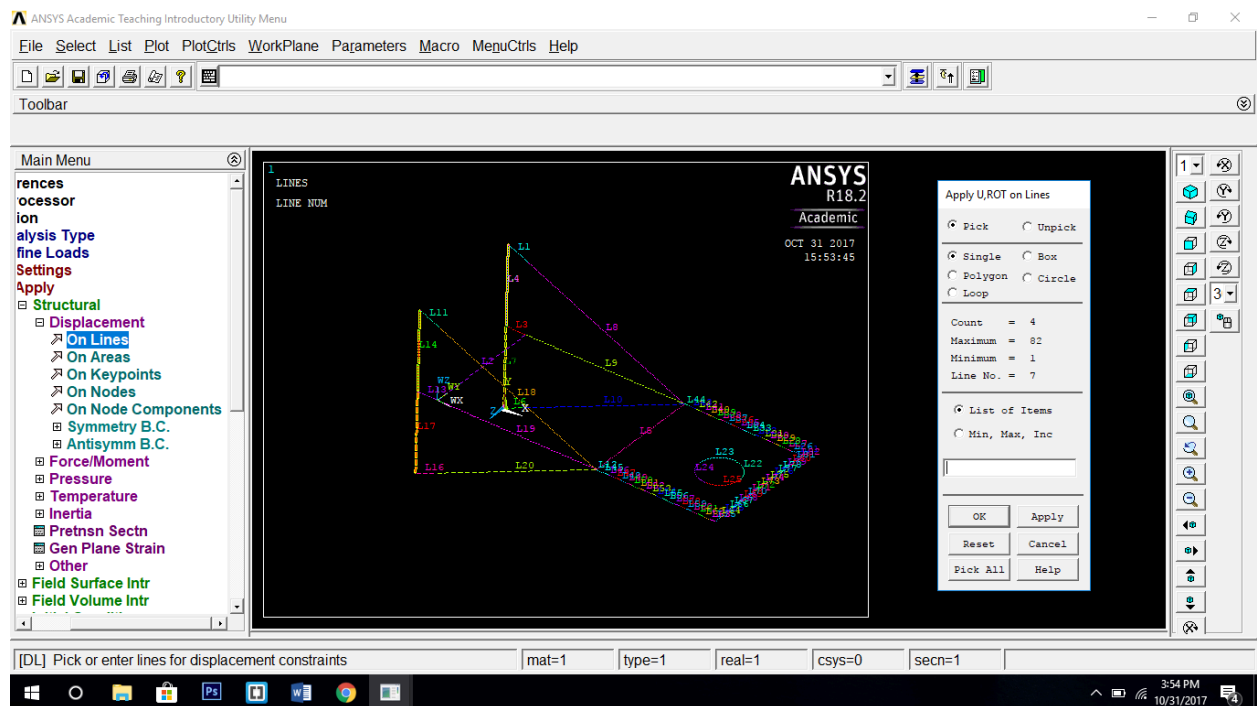


Figure 22 Selecting the lines for Displacement

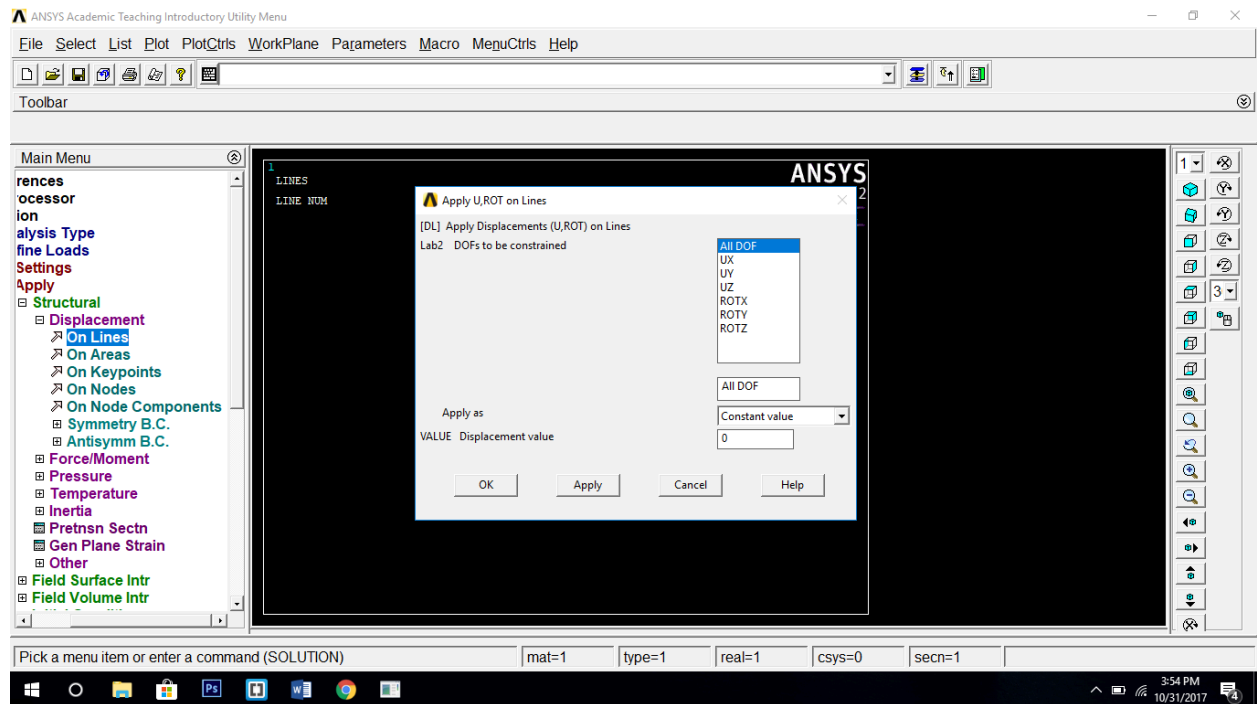


Figure 23 Adding Displacement on the lines (2)

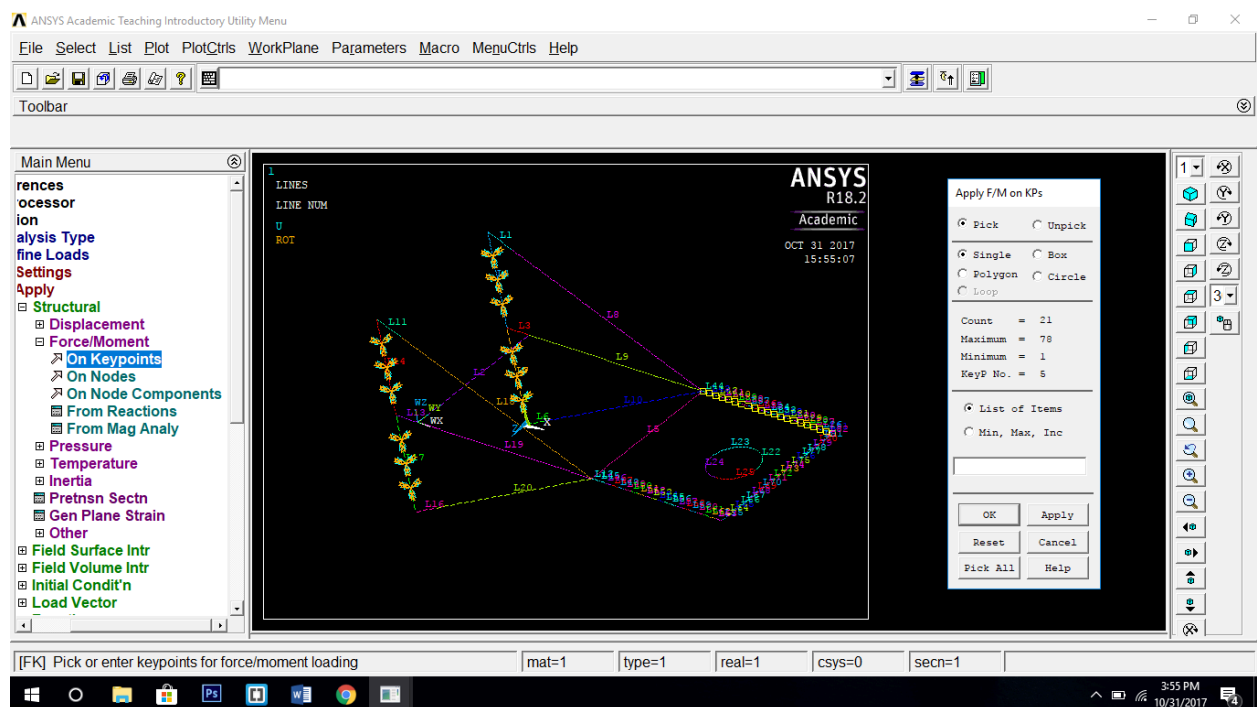


Figure 24 Displacement done at fixed points

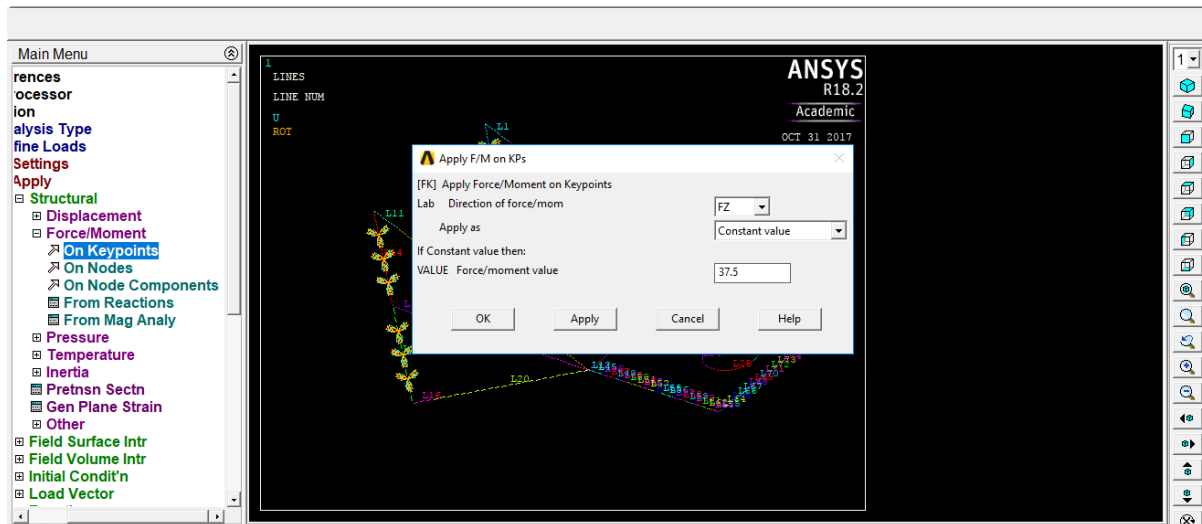


Figure 25 Applying force at Z direction

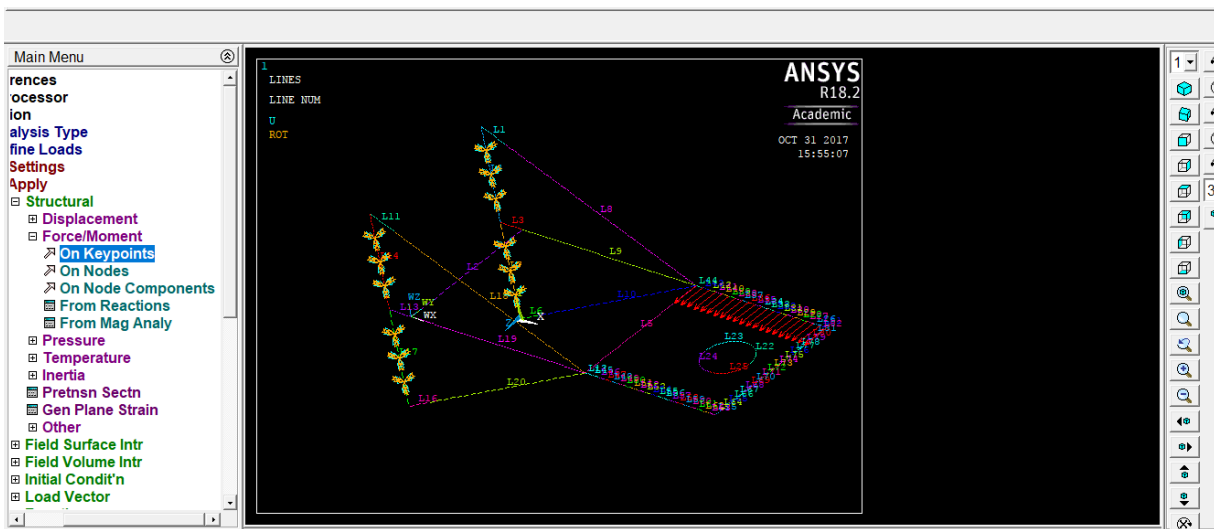


Figure 26 Force at Z-direction

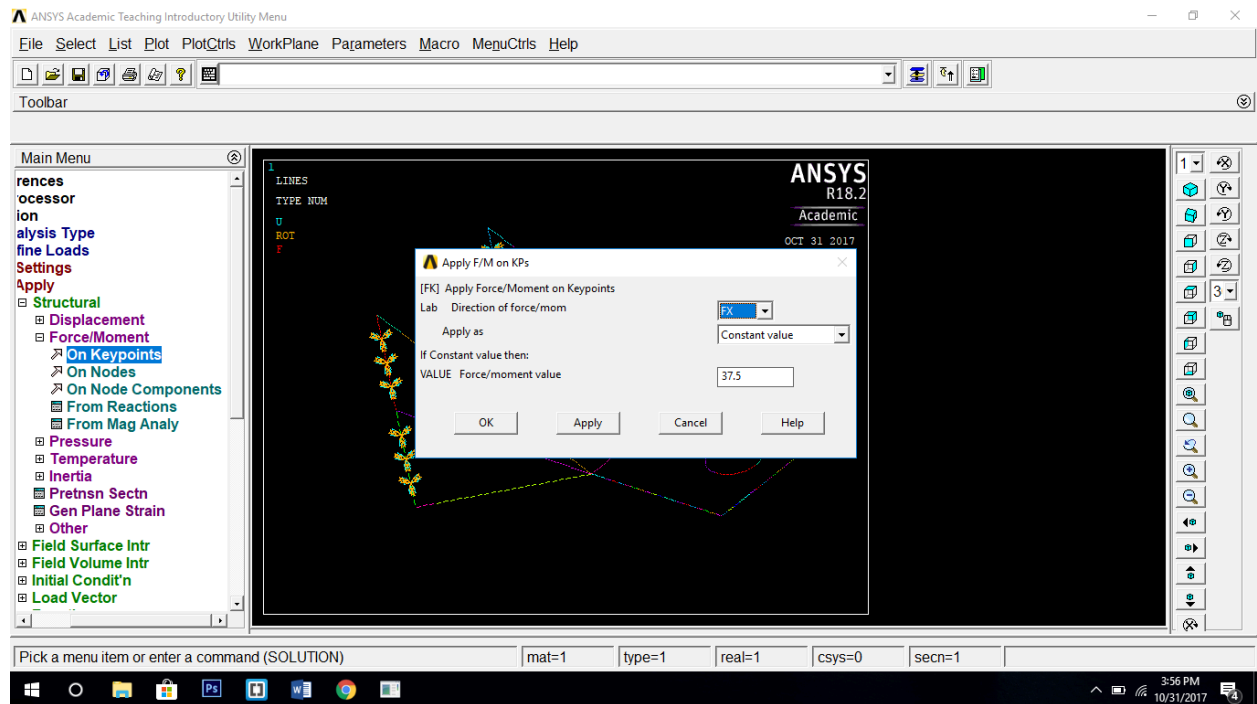


Figure 27 Force at X-direction

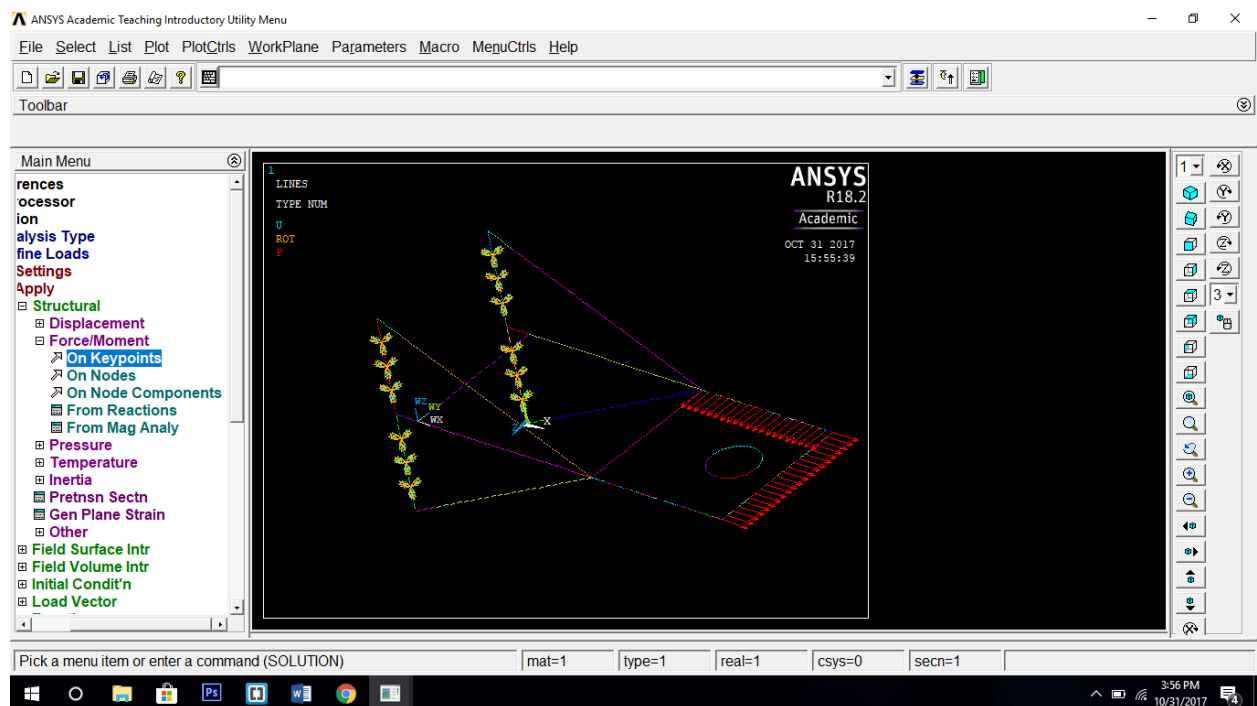


Figure 28 Force Applied at X direction

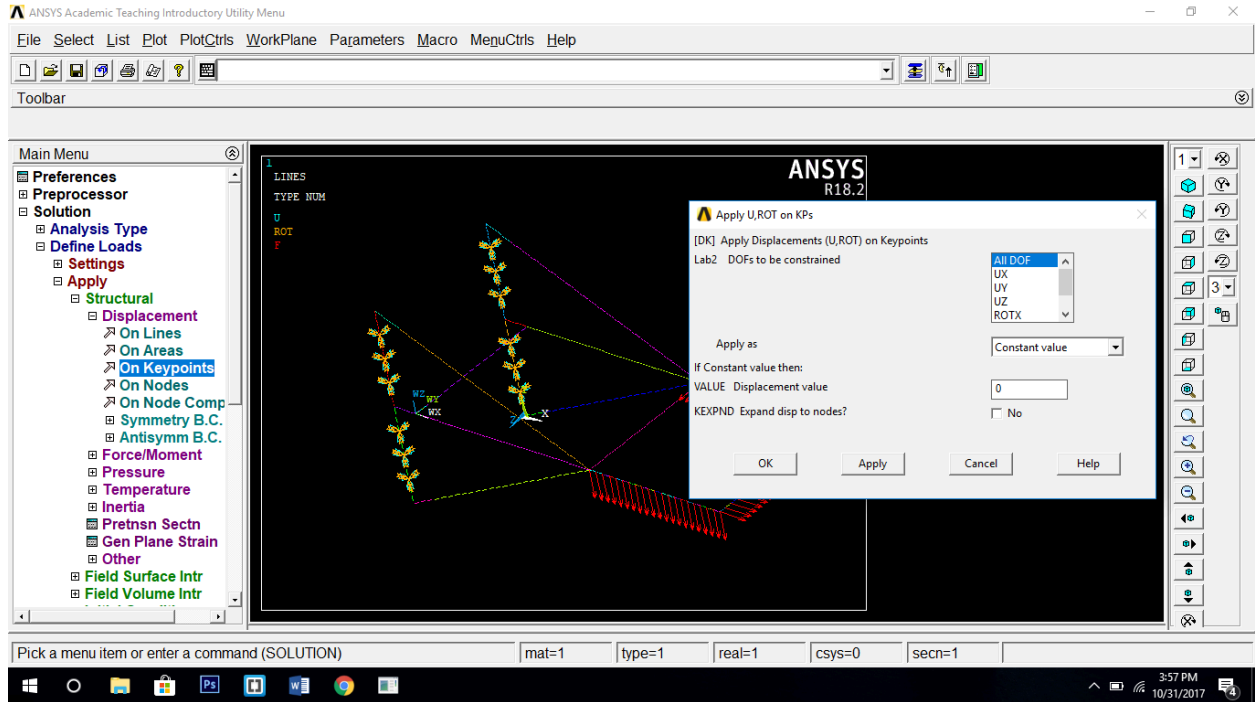


Figure 29 Applying Forces in all three directions

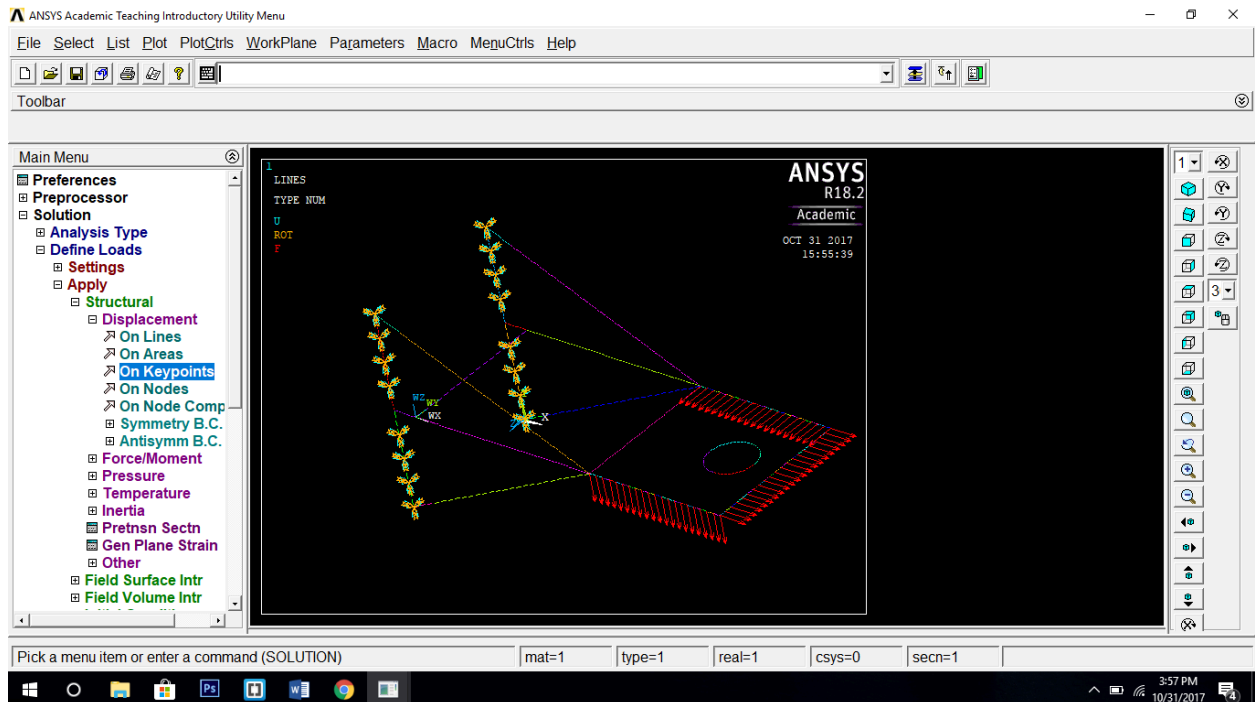


Figure 30 Full Forces in three directions

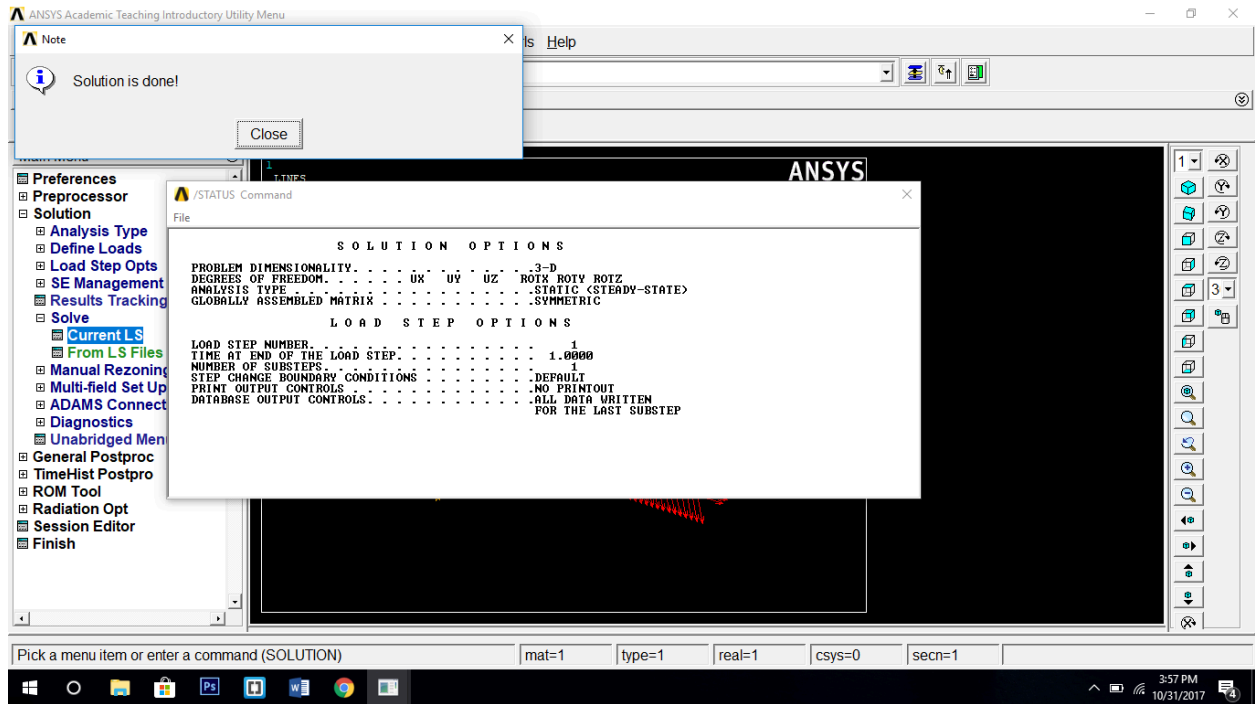


Figure 31 Solving

General Post Processing

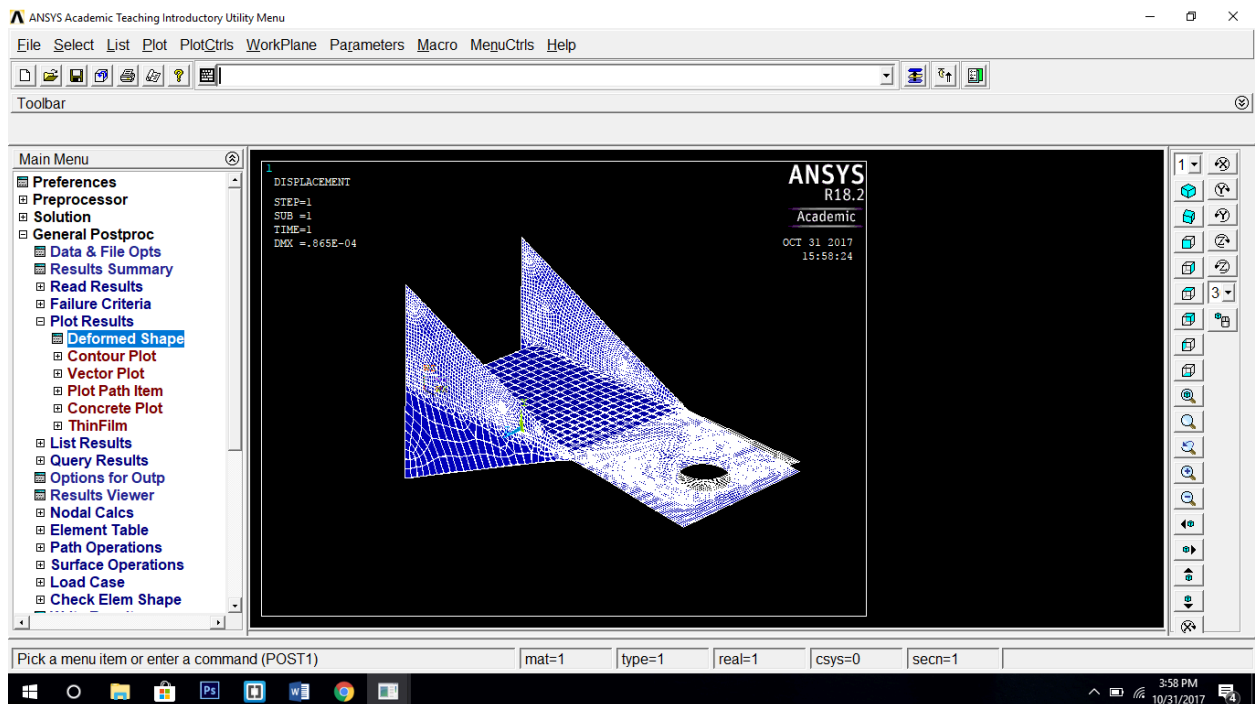


Figure 32 Deformation

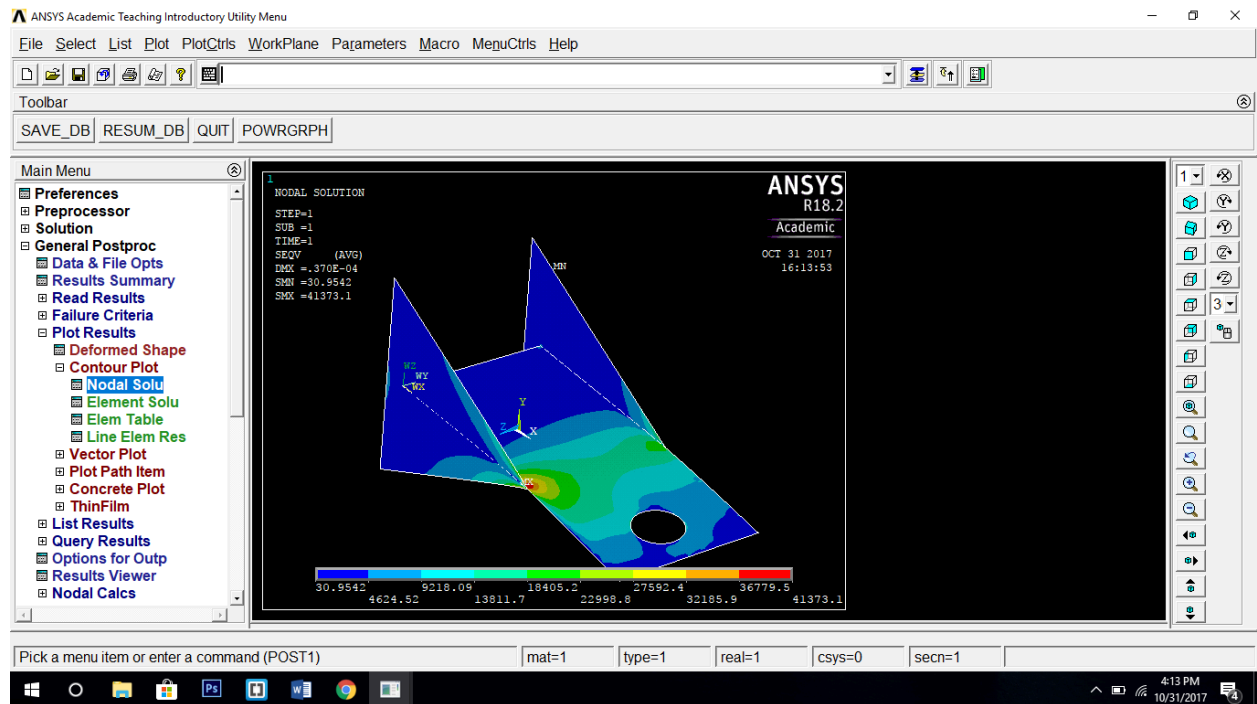


Figure 33 Von Mises Stress

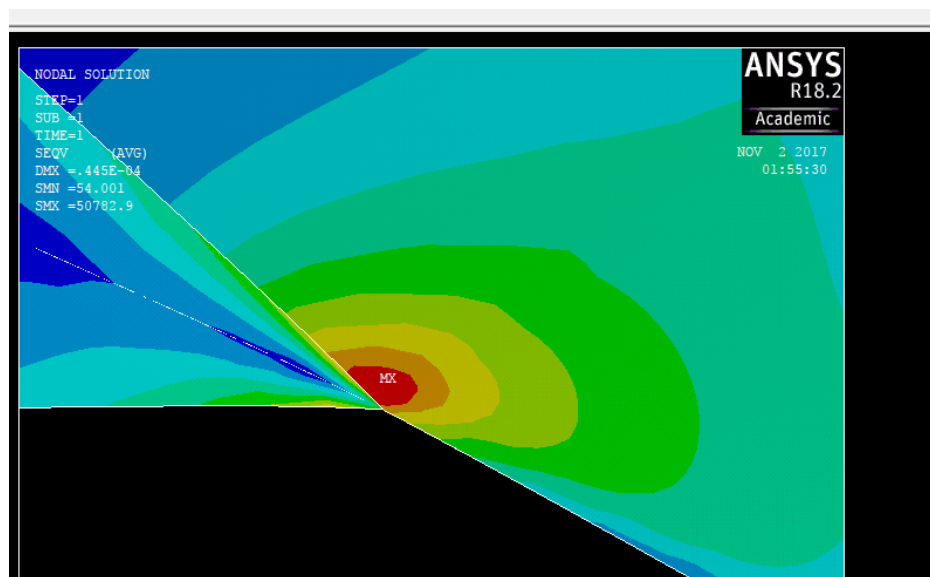


Figure 34 Critical Section

At this point, we have the highest stress acting at this joint. This is due to the Force at the y direction component acting against the welded joints. This causes bending moment around the x-axes causing it to have high stress. The critical stress factor is found by

$$\sigma = \frac{373.2}{50} = 7.5 \text{psi}$$

Model Verification

Case 1

Considering the different types of loading separately

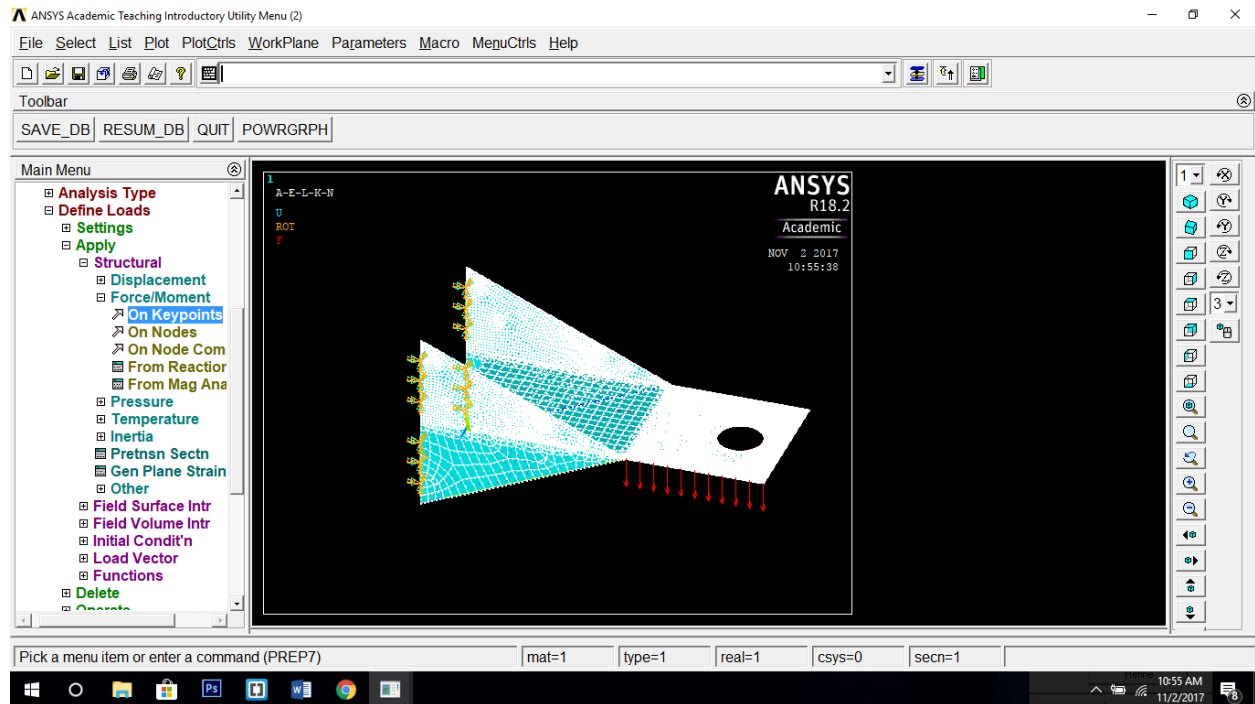


Figure 35 Bending Force being applied

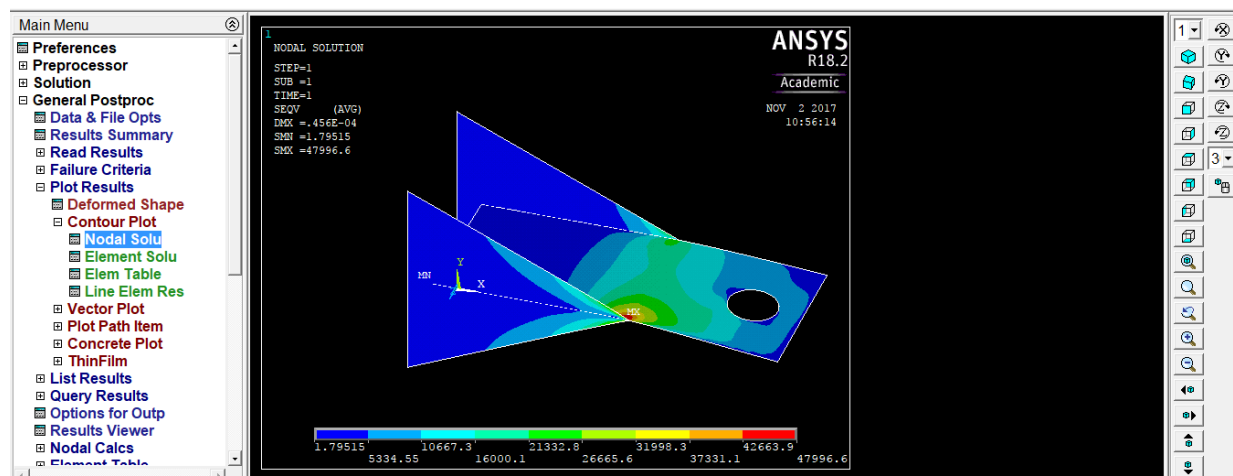


Figure 36 bending Force

Over here we see that there is a critical section occurring due to the force being applied downwards. This causes the bending of the bracket which interns provides more stress at the welded parts. We can have the stress concentration factor at Mx to be the same as our previous calculation.

$$\sigma = \frac{373.2}{50} = 7.5psi$$

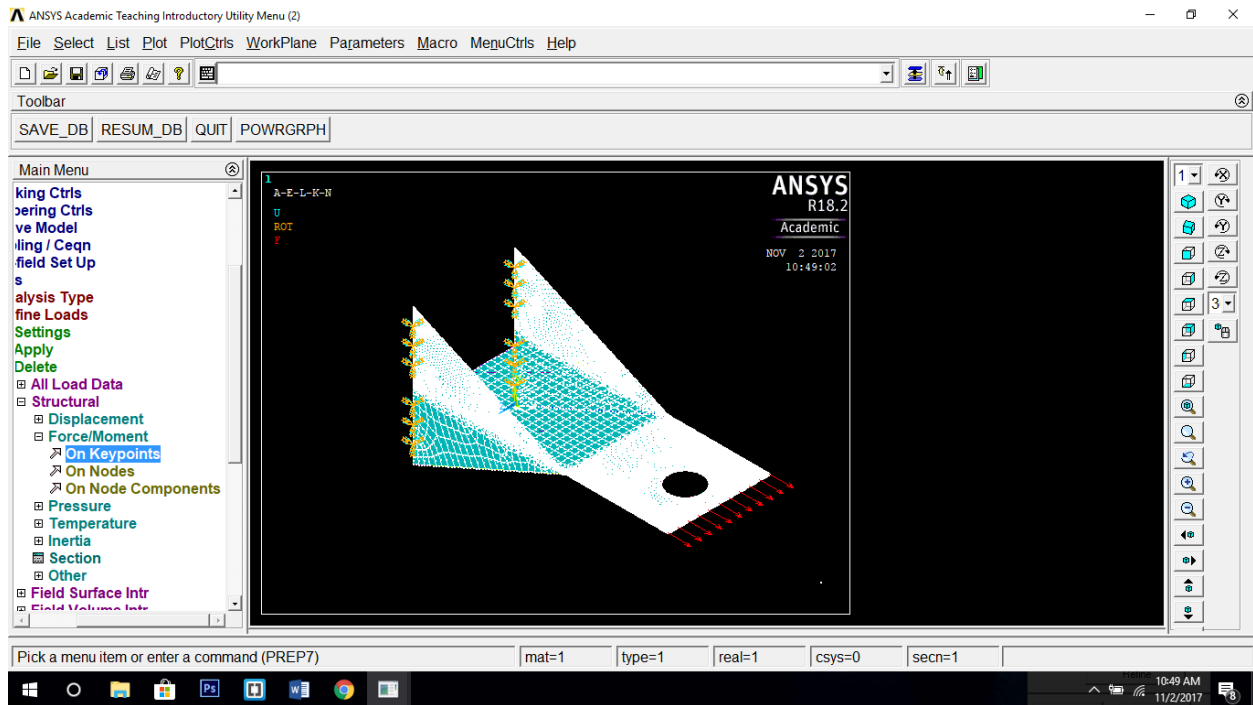


Figure 37 Axial Loading

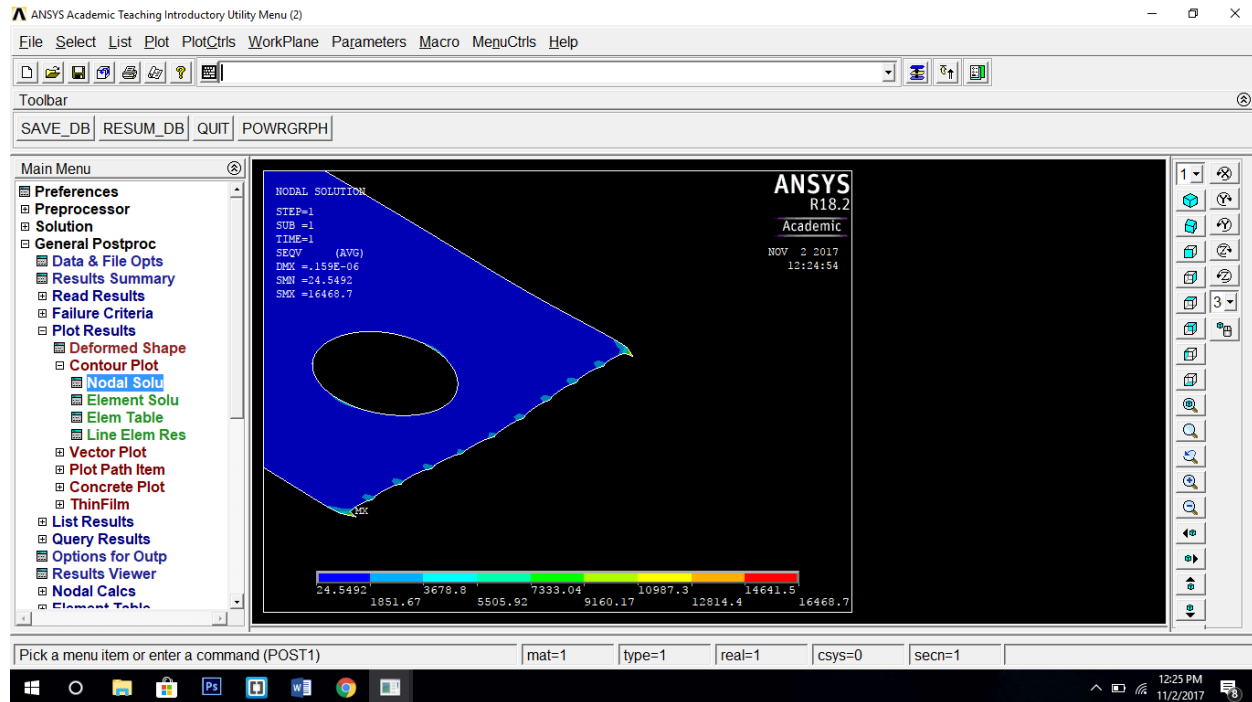


Figure 38 Von misses stress at axial

In the axial loading, we see that the force does not produce any critical sections, it has stress concentration factor of

$$\sigma = \frac{73.3}{50} = 1.467 \text{ psi}$$

From this we can see that only Fy force component is giving out a critical section stress at the welded points. This shows of concern and the causes can be of the following:

- Improper welding of joints
- Lack of fusion also known as cold lapping or cold shuts, this type of welding occurs when the base plate (for us is the rectangular plate with a hole) is not properly welded to the surface (triangular plates)
- Another form of is residual stress occurring at the base plate and these happen at the lowest temperature, moreover; the base plate might show cracking after days. Cold cracking is limited to steel which is our material that we are using for the bracket. (Esab, 2017)

Solution to the cold cracking phenomenon:

- Doing post weld heat treatments
- Reducing the cooling rate by using heat blankets all together
- Selecting low filler hydrogen

Case 2

Modifying the mesh properties for better results at separate loadings.

Axial loading

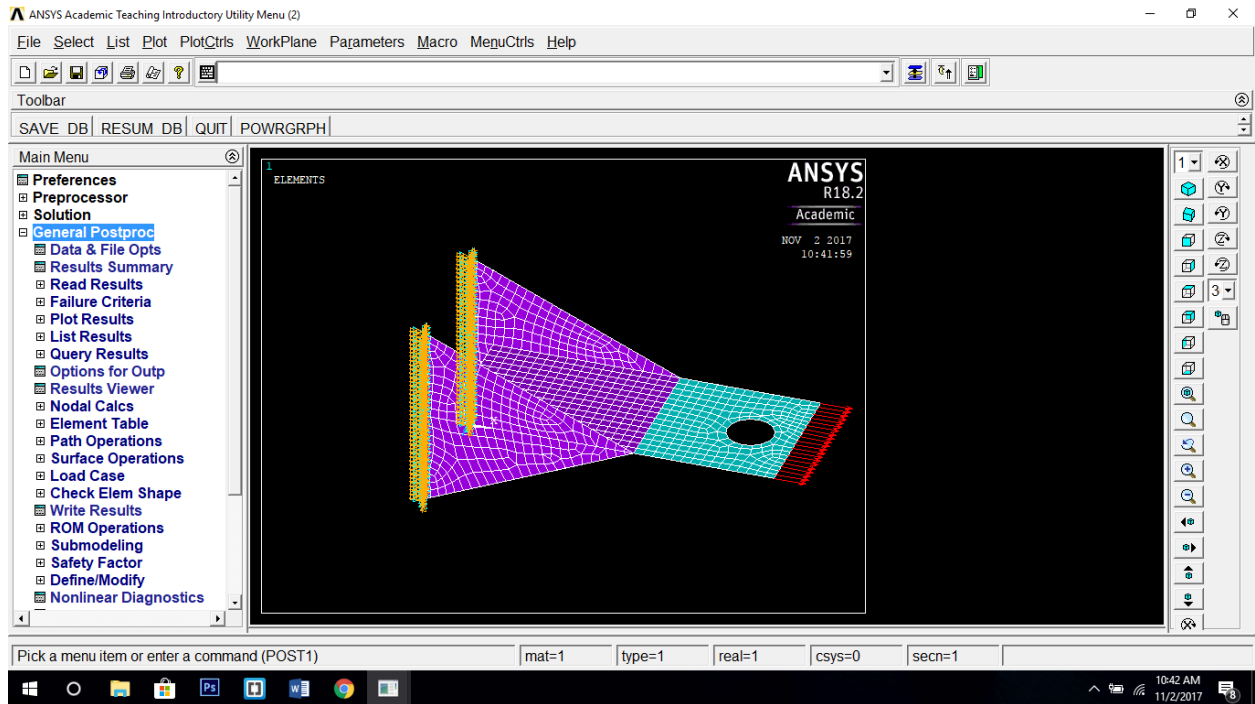


Figure 39 Initial Mesh of Axial Loading

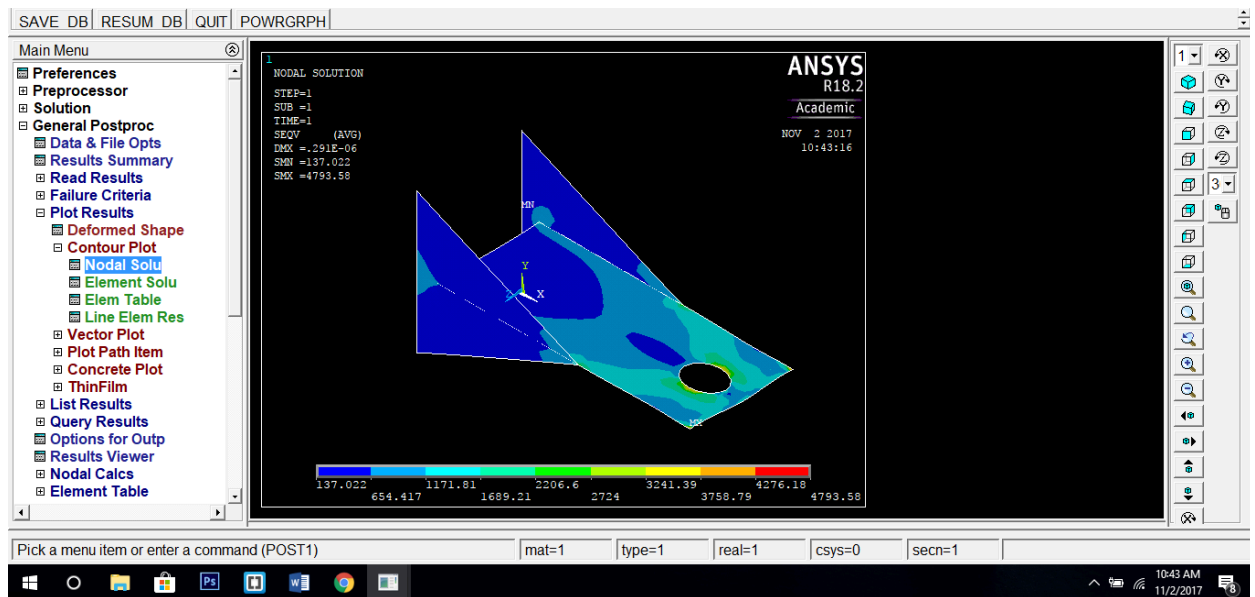


Figure 40 Initial Results of Axial loadings

After meshing

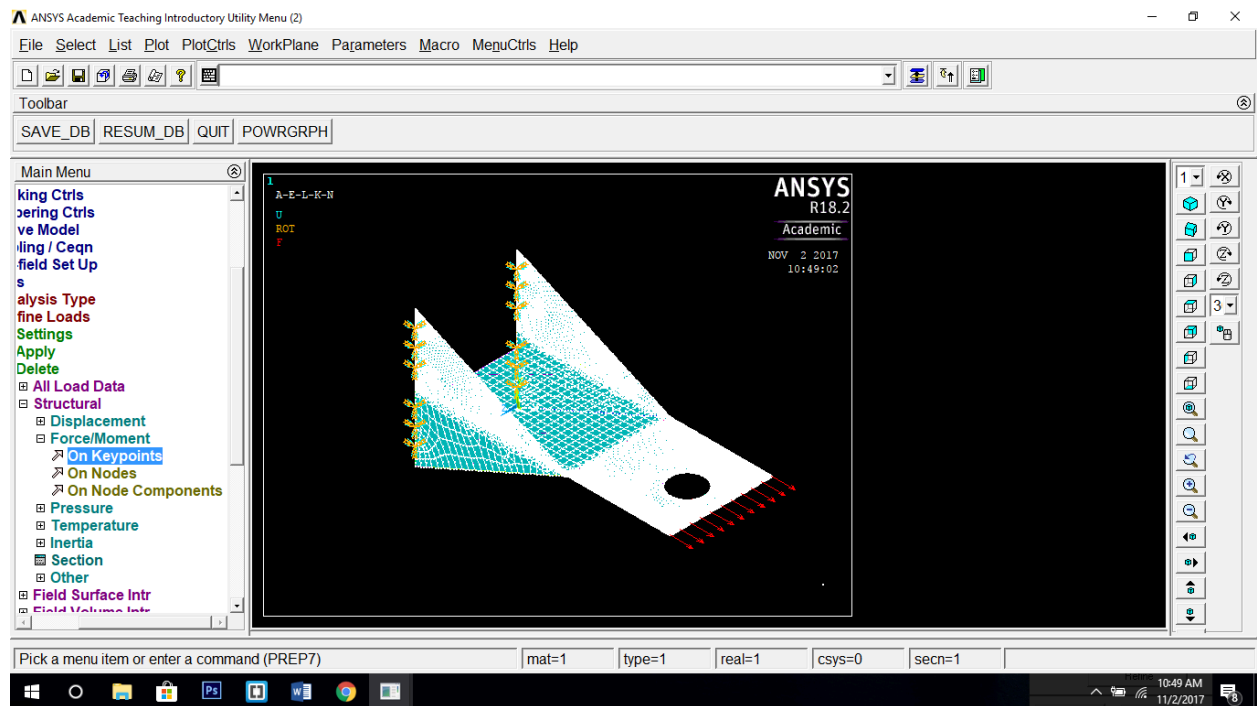


Figure 41 Refined Mesh

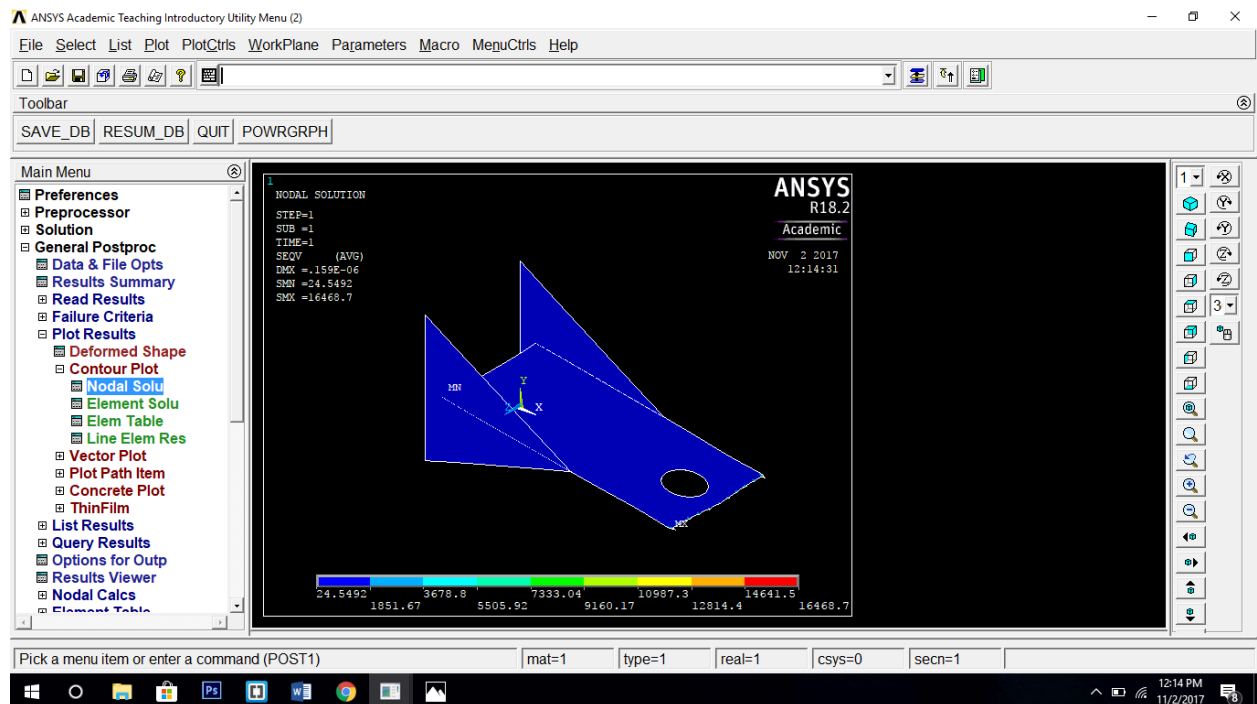


Figure 42 Von Mises of Refined Mesh

In Axial Loading we see that there is a drastic change by refining the mesh. At initial mesh, we see that there is residual stress all over the rectangular plate and even crossing the welding parts. As soon as we refined the mesh quality of the areas where the forces are being applied. We see that there is a huge difference of Von Mises at the lines where forces are at.

At initial mesh, the we have critical stresses around the circle as well as longitude stresses at the edges. The stress concentration factor around the circle is

$$\sigma = \frac{32.43}{50} = 0.65$$

However, after refining the mesh we see that the edges of the rectangular plate are only affected by applying the axial force. This is a logical solution as we can note that the edges will only stretch as tension is being applied to the bracket. Therefore, we see that after refining the mesh, we have a stress concentration factor at edges

$$\sigma = \frac{10.98}{50} = 0.22$$

For Bending Force

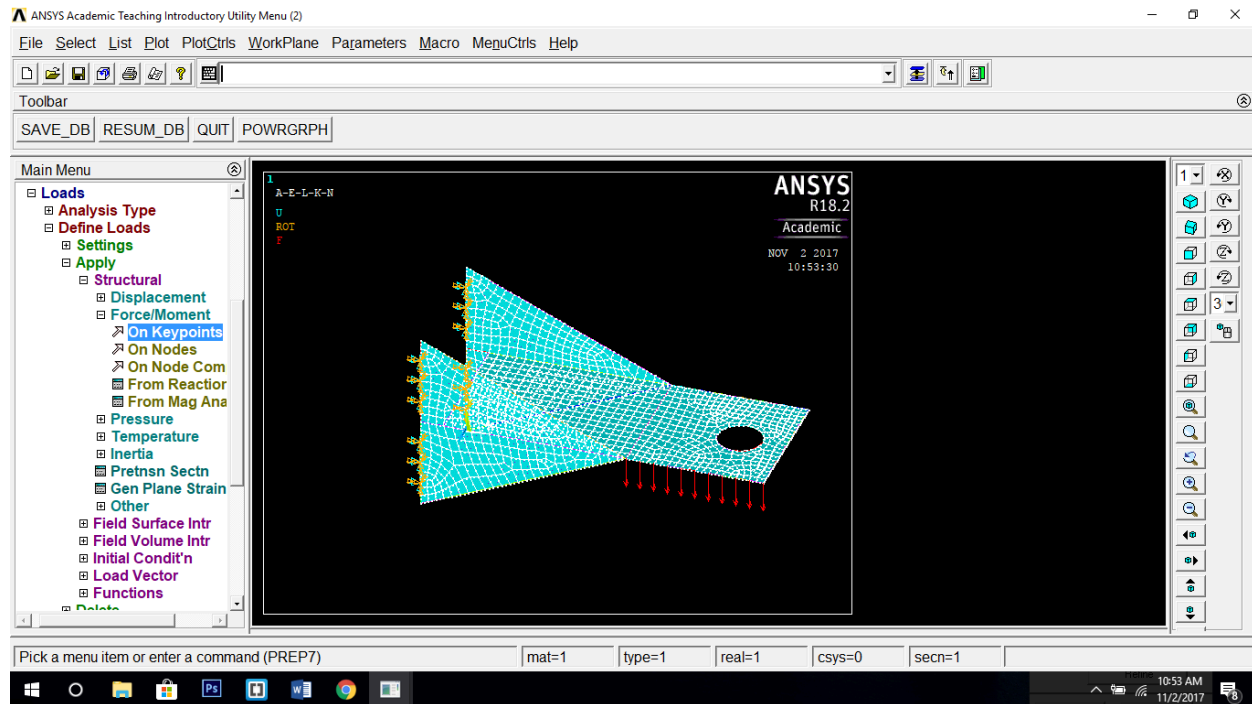


Figure 43 Initial Mesh

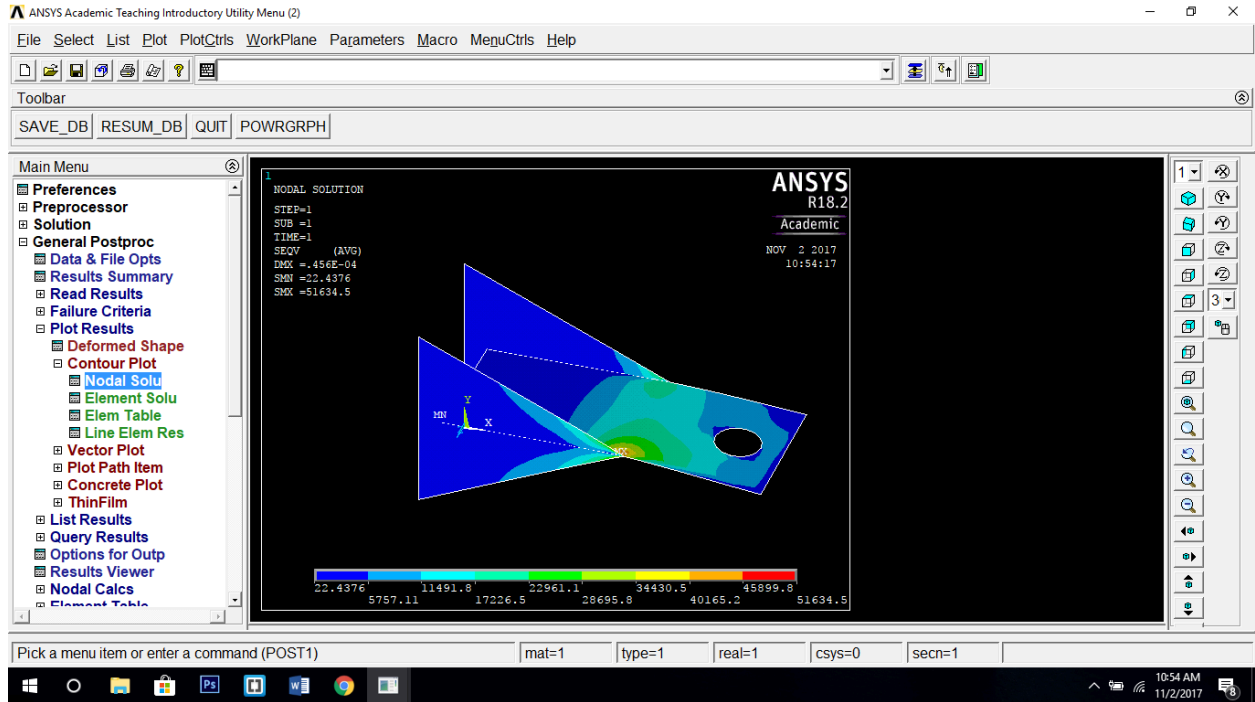


Figure 44 Von Mises

After Meshing

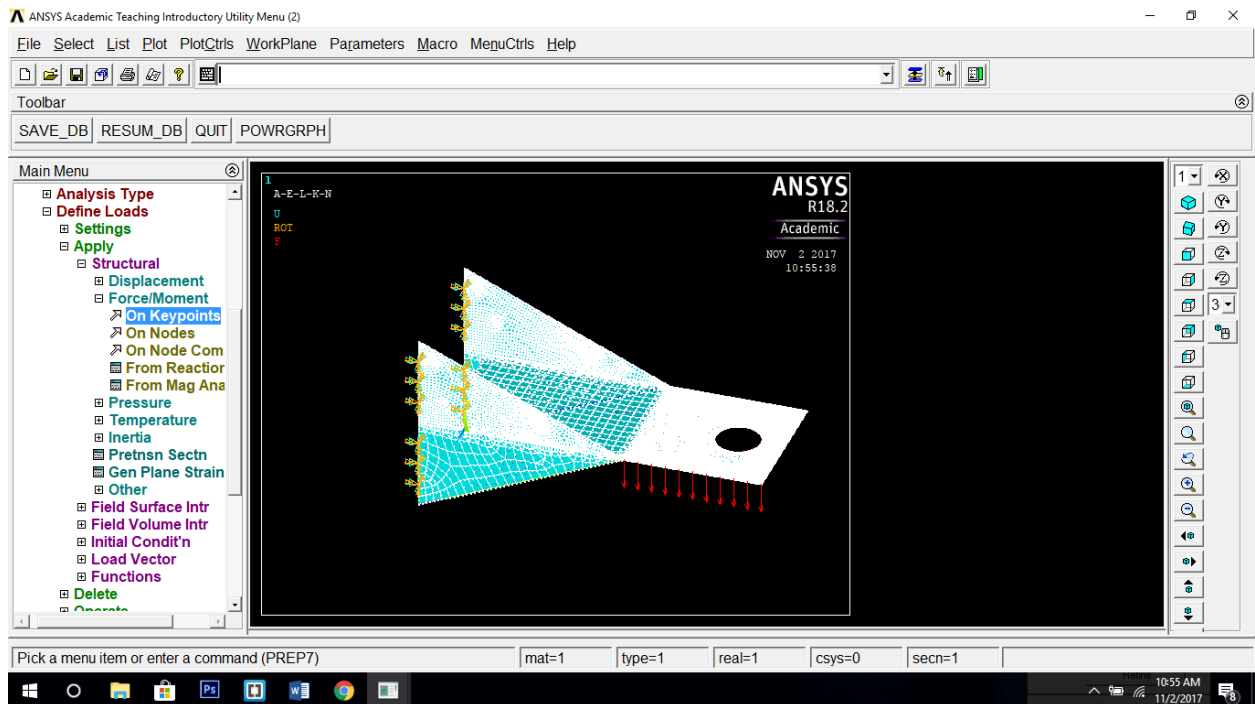


Figure 45 Refined Mesh

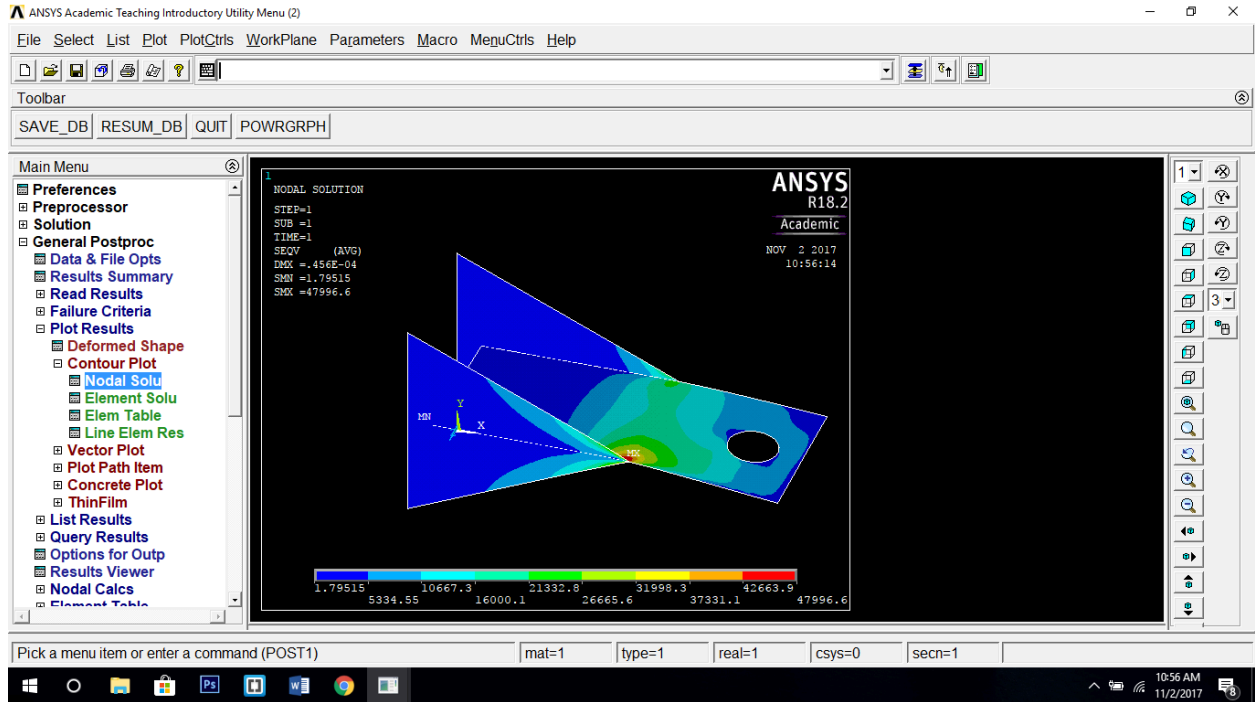


Figure 46 Von Mises

In the bending force, we see that the stresses are critical in both the factors of the meshing. However, as we refine the mesh, the better results we get.

In initial mesh condition, we see that the critical condition is starting to form at 40.16 psi and increase to 46.0 psi.

Using the stress at where the critical stress to occurs we stress concentration factor as

$$\sigma = \frac{40.16}{50} = 0.80$$

After refining the meshing, we see a clear critical section from 37.3 psi to 42.63 psi

$$\sigma = \frac{37.3}{50} = 0.746$$

In short notice, we see that the critical stress concentration factor to occur is 0.746, to avoid critical stress sections, we need to better welding joints or reducing cold cracking.

Case 3

Hand calculation stress concentration factor,

If we are taking only axially and bending forces into a 2D problem.

To calculate the stress factor (axially)

$$K_t = 3.000 - 3.140 \left(\frac{d}{2c} \right) + 3.667 \left(\frac{d}{2c} \right)^2 - 1.527 \left(\frac{d}{2c} \right)^3 \quad (\text{Hartford, 2017})$$

Theoretical Calculation is

$$K_t = 3 - 3.140 \left(\frac{1}{2(1.5)} \right) + 3.667 \left(\frac{1}{2(1.5)} \right)^2 - 1.527 \left(\frac{1}{2(1.5)} \right)^3 \quad (\text{Hartford, 2017})$$

$$K_t = 2.3$$

From Ansys, we got our stress concentration factor (axial) as

$$K_t = \frac{138}{50} = 2.72$$

$$\text{Percentage error} = \frac{\text{Exact} - \text{Approximation}}{\text{Exact}} \times 100$$

$$\text{Percentage error} = \frac{3.2 - 2.3}{3.2} \times 100 = 28.5\%$$

For Bending stress concentration factor

$$K_{tB} = C_1 + C_2 \frac{c}{e} + C_3 \left(\frac{c}{e} \right)^2$$

	$0 \leq d/2c \leq 0.5, \quad 0 \leq c/e \leq 1.0$
C_1	$3.000 - 0.631(d/2c) + 4.007(d/2c)^2$
C_2	$-5.083 + 4.067(d/2c) - 2.795(d/2c)^2$
C_3	$2.114 - 1.682(d/2c) - 0.273(d/2c)^2$

From our Ansys we had our bending stress concentration factor as 0.74

From theoretical we will have

$$C_1 = 3 - 0.631 \left(\frac{1}{3} \right) + 4.007 \left(\frac{1}{3} \right)^2 = 3.23$$

$$C_2 = -5.083 + 4.067 \left(\frac{1}{3} \right) - 2.795 \left(\frac{1}{3} \right)^2 = -4.038$$

$$C_3 = 2.114 - 1.682\left(\frac{1}{3}\right) - 0.273\left(\frac{1}{3}\right)^2 = 1.523$$

$$K = 3.23 - 4.038 + 1.523 = 0.715$$

$$\text{Percentage error} = \frac{0.75-0.72}{0.75} \times 100 = 4\%$$

Conclusion

To conclude the whole Ansys project required the following steps, modelling plan, Ansys steps, model verification to prove that our results are much better. Moreover, some hand calculations were considered only for loading and bending forces.

The results that was obtained in this report were logical and concise, the results included 28.7% and 4% error in comparison of theoretical against numerical analyses.

Taking the 28.5% error into account, they can be many reasons for thermotical not considering such the number of nodes of the element that the force has been applied on. The other factor would be the refining of meshing and its huge element number of sizes. Numerically considers all the factor for calculations while theoretically we are only considering the dimensions of the plane and the force required.

Finally, I would like to conclude this project on the following outcomes achieved

- Practice of the model aspects and turning a real-world problem to a FE model.
- Usage of Ansys Apdl in working the problem.
- Able to verify the results of the problem either by comparison, meshing sizes or hand to hand calculations.
- Practice in depth of Ansys Apdl for other thermal or Fluid related problems.

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